

An Overview and Practical Guide to IRT Measurement Equivalence Analysis

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Abstract

This article provides an overview and guide to implementing item response theory (IRT) measurement equivalence (ME) or differential item functioning (DIF) analysis. We (a) present the need for establishing IRT ME/DIF analysis, (b) discuss the similarities and differences between factor-analytic ME/DIF analysis, (c) review commonly used IRT ME/DIF indices and procedures, (d) provide three illustrations to two recommended IRT procedures, and (e) furnish recommendations for conducting IRT ME/DIF. We conclude by discussing future directions for IRT ME/DIF research.

Keywords

invariance testing, measurement models, item response theory, quantitative research, construct validation procedures, reliability and validity

Assessing the measurement equivalence (ME) of measures used in organizational and psychological research is necessary to ensure that derived scores are comparable across groups (Drasgow, 1987; Vandenberg & Lance, 2000). Analysis of ME in the item response theory (IRT) framework is known as differential item functioning (DIF) analysis. DIF is present when the relationship between the item score and the latent trait is not identical across subpopulations (Drasgow, 1984). Cumulative DIF across all items on a measure can result in differential test functioning (DTF) where the relationship between the test score and the latent trait is not identical across subpopulations. There have been multiple reviews on DIF statistics/procedures (e.g., Penfield & Camilli, 2007; Potenza & Dorans, 1995; Sireci & Rios, 2013; Zumbo, 2007). Yet the diverse and growing number of DIF statistics and procedures has made the issue of conducting DIF analysis challenging for organizational researchers.

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The goal of this article is to provide an overview and practical guide to IRT ME or DIF analysis. We use both terms, *ME* and *DIF*, interchangeably throughout the article, although we primarily use *ME* in the first few sections when discussing the relation between CFA ME and IRT ME. In later sections, we use the term *DIF* in line with the conventional IRT terminology.

The article is structured as follows: We (a) present the rationale for establishing IRT ME procedures within organizational research, (b) discuss similarities and differences between CFA ME and IRT ME, (c) conduct a literature review on IRT ME analysis, (d) provide three illustrations of two recommended IRT ME analysis procedures, and (e) provide a summary of recommendations for conducting IRT ME analysis. We conclude by discussing future directions for IRT ME research.

Rationale for IRT ME

Historically, organizational researchers have received more training on confirmatory factor analysis (CFA) than IRT. As such, early work in the organizational sciences focused on CFA methods of establishing ME, particularly among Likert-type rating scales, culminating with Vandenberg and Lance's (2000) highly cited review of ME analysis. While CFA has served our field well, we propose that IRT provides some advantages over CFA. These include increased model flexibility, provision of useful psychometric statistics/applications, and modeling of more item parameters.

Model Flexibility

Virtually all our psychological and organizational rating scales have discrete categorical response formats (e.g., *strongly disagree* to *strongly agree*). While CFA assumes that the item responses are continuous and linear, IRT models have been developed to model the item responses as either nominal or ordinal variables. This increases our flexibility for modeling different types of data. In the case of nominal variables, common survey categories such as *not sure*, *neutral*, and *?* need not be treated as continuous with some IRT models (Hernandez, Drasgow, & Gonzalez-Roma, 2004). More generally, we can model for nominal responses. In the case of ordinal variables, we do not need to treat rating scale options as having the same fixed intervals. For instance, in a typical 5-point rating scale for *strongly disagree* to *strongly agree*, the distance between 1 = *strongly disagree* and 2 = *disagree* need not be the same as the distance between 2 = *disagree* and 3 = *neither disagree nor agree*. As such, IRT allows researchers to model responses to item options.

With dichotomous items scored as zero or one, CFA methods can lead to expected item responses lower than zero or greater than one with some estimates of ability. However, IRT can be used to specify a nonlinear model so that the expected observed response given the latent trait cannot exceed the bounds of 0 to 1 (Bock, 1997) (see Figure 1A). In the context of multiple-choice test questions that are dichotomously scored (0 = wrong, 1 = right), CFA does not accommodate the propensity to guess on an item, but IRT can be used to model for guessing.

IRT models have been recognized as increasingly important—and arguably better—for handling a variety of measures typically used in organizational research, including attitudes, emotions, personality, and interests (Stark, Chernyshenko, Drasgow, & Williams, 2006; Tay & Drasgow, 2012b; Tay, Drasgow, Rounds, & Williams, 2009). One reason for this is that there are several different IRT models that can accommodate different types of relationships between item responses and the latent trait. For instance, older approaches such as the graded response models are basically extensions of dichotomous models in which higher levels of the latent trait lead to higher observed scores. Conversely, newer IRT models such as ideal point models (Roberts, Donoghue, & Laughlin, 2000) can model non-monotonic relationships between the

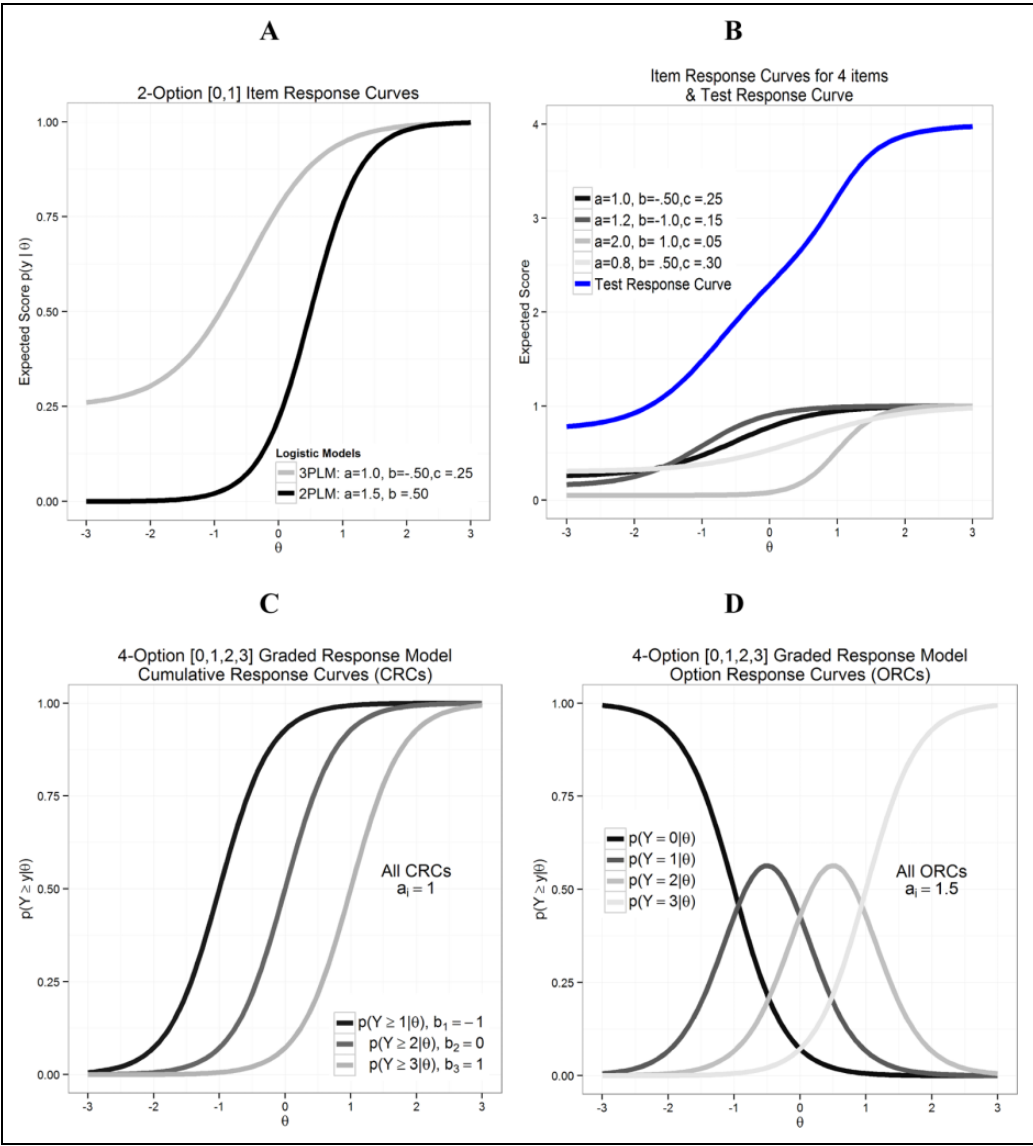


Figure 1. Item response theory plots: two-option and four-option response models. Note: (A) shows the item response curves (IRCs) (expected score as a function of the latent trait) for the 2 parameter logistic model (PLM) and 3PLM; (B) shows the IRCs for four 3PLM items and the test response curve; (C) shows the cumulative response curves (CRCs) (probability of endorsing options greater than or equal to 1, 2, or 3 as a function of the latent trait) for a four-option graded response model; (D) shows the option response curves (ORCs) (probability of endorsing option 0, 1, 2, or 3 as a function of the latent trait) for the four-option graded response model.

latent trait and items responses. For example, under ideal point models, respondents can disagree with moderately worded statements such as “I enjoy the company of a few friends” because they prefer to be alone (lower on the trait) or because they prefer being around many other people (higher on the trait).

Psychometric Statistics/Applications

Within the psychometric literature, it is well known that the traditional conception of reliability (i.e., Cronbach's alpha) erroneously assumes equal accuracy of measurement across the entire range of the latent trait. In contrast, IRT provides estimates of both item and test *information*, which is an indication of score accuracy that varies across levels of the latent trait (Drasgow, Nye, & Tay, 2010). As such, IRT item information can be used for developing tests with maximal accuracy at a given point on the latent continuum (e.g., for a test with a given cut-score) or assembling tests optimized for individual examinees (e.g., computerized adaptive testing). Item differences in precision of measurement across different areas of the latent trait allow for the use of computerized adaptive testing (CAT) and optimal test/scale assembly (Embretson & Reise, 2000). For example, relatively high-ability examinees will be given relatively more difficult items as those items will tend to provide more precision at higher levels of the latent trait. This approach is useful for providing short but accurate selection tests.

Model Parameters

For polytomous data, CFA typically produces a single item intercept per item because it uses a continuous model. In contrast, IRT typically yields multiple parameters analogous to item intercepts per item because IRT models the polytomous data as categorical. Often the CFA and IRT approaches to ME testing yield similar results, particularly when there are mean differences across groups resulting in uniformly higher item parameters in one group than another (Stark, Chernyshenko, & Drasgow, 2006). However, the additional parameters per item afforded by IRT methods can result in more sensitivity to nuanced group differences such as differences in central tendency or the use of extreme scores (Meade & Lautenschlager, 2004). This is because IRT models each response option, unlike CFA.

Summary

IRT can be applied to many types of data (e.g., ability tests, Likert-type rating scales, and items with unordered response options). It is also used to develop more optimal tests (e.g., CAT). Moreover, by typically providing more parameters per item, more information can be gleaned from IRT models than from most CFA models, allowing for more sensitive tests of ME. Despite these advantages, there are three main barriers to utilizing IRT analyses in organizational research. First, organizational researchers tend to be better trained on CFA methods but not in IRT methods. Second, the wide variety of IRT models (and IRT ME indices/procedures) that afford flexibility can be confusing to newcomers. Third, IRT software historically has been notoriously difficult to use. We attempt to address these barriers in the following sections.

IRT ME Versus CFA ME: Parameters, Concepts/Procedures, and Practical Usage

In this section, we seek to provide a broad understanding of the underlying similarities and differences in the parameters, concepts/procedures, and practical usage between IRT and CFA. We provide a framework for IRT ME in light of CFA ME analysis.

CFA and Dichotomous IRT Parameters

In Table 1, we provide an overview of the correspondence between CFA and dichotomous IRT models. Generally speaking, dichotomous IRT a_i parameters are conceptually similar to CFA

Table 1. Correspondence between confirmatory factor analysis (CFA) and dichotomous item response theory (IRT) parameters.

CFA Model	Form 1	Form 2	Definition of Terms and Comments
Single factor FA	Continuous outcome: $y_{ij} = v_i + \lambda_i \theta_j + \varepsilon_{ij}$	Binary outcome: $y_{ij}^* = v_i + \lambda_i \theta_j + \varepsilon_{ij}$ $y_{ij} \begin{cases} 1 & \text{if } y_{ij}^* \geq \tau_i \\ 0 & \text{if } y_{ij}^* < \tau_i \end{cases}$ v_i assumed to be 0	To incorporate binary outcomes, y_{ij} becomes y_{ij}^* and is defined as a <i>latent response</i> or <i>item propensity</i> ; y_{ij}^* produces a value of 1 when it is equal or higher to the item threshold τ_i and a value of 0 otherwise. i and j index the item and the individual, respectively; y_{ij} is the observed response; v_i is the item intercept; λ_i is the item loading; θ_j is the latent trait; ε_{ij} is the residual term; y_{ij}^* is the latent response or item propensity; τ_i is the item threshold
IRT Model	Form 1	Form 2	Definition of Terms and Comments
Normal cumulative distribution function (Φ)	$\Phi(a_i \theta + \beta_i)$ Conversions: $a_i = \frac{\lambda_i}{\sqrt{1-\lambda_i^2}}$ $\beta_i = -\frac{\tau_i}{\sqrt{1-\lambda_i^2}}$ $\beta_i = -a_i b_i$	$\Phi(a_i [\theta - b_i])$ Conversions: $a_i = \frac{\lambda_i}{\sqrt{1-\lambda_i^2}}$ $b_i = \frac{\tau_i}{\lambda_i}$; $b_i = -\frac{\beta_i}{a_i}$	a_i denotes the item discrimination parameter; β_i denotes the item location parameter where larger positive values indicate item easiness; b_i denotes the item difficulty. Note: For items with multiple options (>2), there are multiple b_i s, and these are referred to as item thresholds.
Logistic function	$\frac{1}{1+\exp[-1.702(a_i \theta + \beta_i)]}$	$\frac{1}{1+\exp[-1.702a_i(\theta - b_i)]}$	See above

factor loadings while dichotomous IRT b_i parameters are akin to CFA item intercepts. As shown in Table 1, the CFA model can be described mathematically in two forms, one for a continuous outcome and another for a binary outcome. For the binary CFA model, past work has shown the equivalence of the standardized CFA and a two-parameter IRT model under some parameterizations (Lord & Novick, 1968; Takane & De Leeuw, 1987). However, CFA tests of invariance in organizational research almost always use the continuous parameterization (see Vandenberg & Lance, 2000).

On a more technical note, there are multiple ways of parameterizing the two-parameter IRT model as shown in Table 1. While technical, the parameterization is important to consider because in the practical implementation of IRT ME, different software may use different parameterizations. Specifically, the two-parameter IRT model can be parameterized as a cumulative normal ogive or logistic function as shown in Table 1. The logistic parameterization closely approximates the normal ogive parameterization when a scaling factor of 1.702 is included (see Table 1). Both IRT parameterizations have item discrimination a_i and item difficulty b_i (or item location/easiness β_i) parameters that can be converted from CFA parameters. We note that because the logistic parameterization is more commonly used, we use this parameterization throughout this article.

In general, we recommend that researchers report the IRT model formulation because the magnitude and sign of the parameters may be different depending on the type of IRT model used. This would affect substantive interpretations such as whether an item is easy or difficult and whether an item is discriminating or not.

Table 2. Concepts/procedures associated with confirmatory factor analysis (CFA) and item response theory (IRT).

	CFA	IRT
Types of measurement equivalence (ME)	(Step 1) Configural (Form). The pattern of zero and nonzero loadings is equivalent across groups. This is generally examined in multidimensional constructs. For unidimensional constructs, the goal is to assess if all the items load on a single dimension across groups. (Step 2) Metric (Loadings/Slopes): The magnitudes of factor loadings are equivalent across groups. This implies that item-level slopes are equivalent. $\lambda_{i, \text{Group1}} = \lambda_{i, \text{Group2}}$ (Step 3) Scalar (Intercepts/Thresholds). The magnitudes of item-level intercepts are equivalent across groups. $\nu_{i, \text{Group1}} = \nu_{i, \text{Group2}}$ (Step 4: Optional) Uniqueness (Error). The magnitudes of item-level uniquenesses are equivalent across groups. The implication is that scale reliabilities are equivalent, but <i>only if</i> factor variances are equivalent. Because there is no a priori reason why factor variances should be equivalent, this step is optional. $\varepsilon_{i, \text{Group1}} = \varepsilon_{i, \text{Group2}}$	(Step 1) Unidimensionality. There is sufficient unidimensionality across groups. Finding unidimensionality across groups implies that there is configural ME. Local independence. After controlling for latent trait, the observed responses are uncorrelated. Nonuniform differential item functioning (DIF). The magnitude of focal item discrimination is not equivalent across groups. $a_{i, \text{Group1}} \neq a_{i, \text{Group2}}$ Uniform DIF. The magnitude of focal item location/difficulty is not equivalent across groups. $b_{i, \text{Group1}} \neq b_{i, \text{Group2}} \text{ OR } \beta_{i, \text{Group1}} \neq \beta_{i, \text{Group2}}$ For items with multiple options (>2), multiple b_i s for an item will be compared simultaneously across groups. In nonlinear IRT measurement models, the uniquenesses are modeled without explicit parameterization. There is no comparable IRT ME concept.
Associated concepts	Complete/Full. The concept of complete/full ME is that <i>all</i> the scale items exhibit a type of ME (i.e., configural, metric, scalar, or uniqueness). Partial ME. The concept of partial ME is that <i>some</i> but not all the scale items exhibit a type of ME (i.e., configural, metric, scalar, or uniqueness).	Differential test functioning (DTF). Differential test functioning indicates that the scale/test as a whole is not equivalent across groups. DTF may be evaluated by the area between test response curves (see Figure 1B for an example of a test response curve). DIF. Differential item functioning indicates that a specific item of interest is not equivalent across groups. DIF may be evaluated not only by equivalence of item parameters, but also other methods (e.g., area between item response curves).
Approaches to testing ME for specific items	Free-baseline. Same concept as IRT.	Free-baseline. Assuming configural equivalence, the free-baseline approach uses a baseline model with item parameters that are freely estimated across groups. Subsequent models constrain item parameters to

(continued)

Table 2. (continued)

	CFA	IRT
	Free-baseline reference indicator. As a default practice, the first item on the scale is usually set to be equivalent across groups and equal to 1.0 in order to provide scaling for the underlying factor. This assumes that the first item is equivalent. It is also possible to have multiple anchor items if invariant items are known a priori.	be equivalent across groups. Comparisons of fit between baseline and subsequent models are made.
	Constrained baseline. Same concept as IRT.	Anchor/Linking items. There are some items that are known to be equivalent across examined groups. These are set as equivalent across groups for the analysis. This is analogous to the free-baseline reference indicator in CFA where the first item is assumed to be equivalent.
Population heterogeneity equivalence or structural equivalence	Factor variances. The factor variances are equivalent across groups.	Constrained baseline. Assuming configural equivalence, the constrained baseline approach uses a baseline model with all item parameters that are constrained as equal across groups. Subsequent models free specific item parameters to be equivalent across groups. Comparisons of fit between baseline and subsequent models are made.
	Factor covariances. The factor covariances are equivalent across groups.	Latent variance. The latent variances are equivalent across groups.
	Factor means. The factor means are equivalent across groups.	Latent covariances. Not possible with unidimensional IRT.
		Latent means. The latent means are equivalent across groups.

CFA and IRT ME Analysis Concepts/Procedures

As shown in Table 2, there are both similarities and differences between CFA and IRT in the types of ME (e.g., configural, metric, scalar, uniqueness), approaches for testing item ME (e.g., free-baseline, constrained baseline), and testing for structural equivalences (e.g., latent means, latent covariances). Differences in CFA and IRT models arise because the substantive aims for CFA and IRT ME have traditionally been different. In CFA ME analysis, the goal is often to demonstrate that group observed scores (i.e., scores for the entire scale) are comparable so that substantive interpretations on observed group differences can be made. CFA ME procedures are frequently treated as a prerequisite for comparing group mean differences on a variable of interest. There have also been attempts to use CFA ME to demonstrate equivalence of construct interpretation across groups (e.g., Mencl, Tay, Schwoerer, & Drasgow, 2012; Riordan & Vandenberg, 1994) where the general form of ME (i.e., configural, metric, scalar, uniqueness) has substantive meaning (Millsap & Hartog, 1988; Thompson & Hunt, 1996). Nevertheless, the focus with CFA is usually at the level of construct construal rather the functioning of individual items.

Unlike CFA ME, IRT ME procedures were developed to assess the ME (or more specifically the lack of ME) for items, hence the terminology differential *item* functioning. This is because IRT was developed in the testing context with a focus on identifying specific test items that may exhibit bias

against a particular group (particularly important for item banking). This is not to suggest that test/scale score equivalence is unimportant, as one can determine the extent to which DIF items cumulatively impact observed mean differences across groups (i.e., differential test functioning [DTF]; where the test response curve shown in Figure 1B is different between groups). Because of the pragmatic concerns of test development in IRT, there is also less emphasis on differences in construct construal between groups when nonequivalence is found. Rather, DIF is often used to identify potential issues in item content and wording (e.g., Ellis, 1989). As such, the differential emphasis between IRT and CFA on items and constructs, respectively, has led to divergences in concepts/procedures though there are underlying similarities.

One notable difference is in the typical steps of ME analysis. ME in CFA is generally tested in a step-wise fashion, assessing types of equivalence from configural, metric, scalar, to uniqueness (Vandenberg & Lance, 2000). At each step, greater constraints are placed on a model, and if fit is significantly worse, it indicates that the focal type of equivalence for the entire scale is lacking (Cheung & Rensvold, 2002). In this procedure, the types of equivalences are emphasized and tied to substantive meaning (Millsap & Hartog, 1988; Thompson & Hunt, 1996). For example, a lack of configural equivalence would indicate that the construct of interest has fundamentally different definitions across groups.

In IRT, assessing unidimensionality corresponds to the first step of configural ME in CFA. It is less common (though possible due to parameter equivalences for some IRT models) to implement the same CFA steps because the goal is often to determine whether specific items have DIF. To test for IRT ME, an item-level analysis is usually undertaken. This often occurs through a comparison of IRT models using either the free-baseline or fully constrained baseline approach (see Table 2) (Stark, Chernyshenko, & Drasgow, 2006). We note that the free-baseline and fully constrained baseline approaches to testing *individual* items for nonequivalence has been shown to be applicable in a CFA ME framework as well. Nevertheless, this is usually less common in a CFA ME framework.

A second significant difference is that although with IRT it is possible to test for different forms of item-level DIF, the presence of DIF rather than the specific nature of DIF has usually been the focus. This is not to suggest that the form of DIF is unimportant. Instead, the goal is usually to determine whether DIF leads to DTF that has a substantial effect on test score comparison (e.g., Stark et al., 2004). Nonuniform DIF may result in no impact on observed scores as lower scores for the focal group (as compared to the reference group) on a given latent trait level may be compensated for by higher scores on another trait level. Further, in a number of IRT procedures, there is generally lower power to detect nonuniform DIF (than uniform DIF; e.g., Finch & French, 2007; Narayanan & Swaminathan, 1996; Swaminathan & Rogers, 1990; Tay, Vermunt, & Wang, 2013), making it more difficult to draw valid inferences about the specific form of DIF. This is juxtaposed with the inherent substantive interest in the form of nonequivalence in CFA ME (e.g., metric equivalence or scalar equivalence).

A third difference is that with the requirement of unidimensionality for most IRT models as currently used, it is impossible to assess the ME of relationships among constructs in the same way as for CFA. Although newer multidimensional IRT models have been developed (Reckase, 2009), these are beyond the scope of the current article and have been rarely employed in organizational research.

A final difference is that in CFA ME analysis, general model-data fit indices for structural equation modeling (SEM) are often used. These include comparative fit index (CFI), non-normed fit index (NNFI or Tucker-Lewis index [TLI]), and root mean square error of approximation (RMSEA) (e.g., Cheung & Rensvold, 1999, 2002). On the other hand, item-level indices for detecting DIF are used extensively in IRT ME analysis. These item indices are broader than testing the equivalences of item parameters (e.g., Lord, 1980; Thissen, Steinberg, & Wainer, 1993) and have included (not exhaustively): area between item response curves (Raju, 1988), logistic regression (Swaminathan & Rogers, 1990), and observed score differences (Holland & Thayer, 1988; Narayanan & Swaminathan, 1994).

Table 3. Research question and measurement equivalence approaches.

Equivalence Type	Example Question	Best Approach	Why?
Relationship among factors	Are the latent trait factors correlated in the same way across groups?	Confirmatory factor analysis (CFA)	Item response theory measurement equivalence (IRT ME) procedures are generally used for unidimensional scales. IRT ME analysis for multidimensional models are less established. In a multidimensional case, IRT ME analysis is usually applied to each dimension separately.
General equivalence of scale scores	Is the scale (score) comparable across groups?	Depends	Both IRT and CFA can be used to answer this question. It depends on whether one desires to assume that the indicators are continuous, and hence CFA is more appropriate, or if indicators are categorical, IRT is more appropriate.
Equivalence of scale items	Are individual scale items comparable across groups?	Depends, but IRT is preferable	Both CFA and IRT can be used to answer this question. However, if individual items are of interest, somewhat more information is gained via the additional item parameters in IRT. Moreover, item-level DIF tests and effect sizes are readily available.
General equivalence of test scores	Is the test (score) comparable across groups?	IRT	IRT procedures are preferable as they can model for "guessing."
Equivalence of test items	Are individual test items comparable across groups?	IRT	IRT procedures are preferable as they can model for "guessing."

Practical Usage of CFA and IRT ME

While the standardized binary CFA and a two-parameter IRT model are mathematically equivalent, there continue to be differences in the practical usage of CFA and IRT ME for several reasons: (a) It is difficult to use multidimensional IRT models for ME analysis whereas CFA models can be easily applied to ME analysis of multiple factors; (b) unlike CFA models, IRT models have been developed to incorporate guessing, which is more appropriate for multiple choice test data; (c) IRT models can estimate multiple response indicator thresholds whereas CFA models do not, producing greater resolution in understanding item responding. As shown in Figures 1C and 1D, responses to multiple options can be modeled for items. Based on this, as shown in Table 3, we present when and why CFA or IRT may be more appropriate for different types of measurement equivalence issues and questions. We note that the term *scale* refers to psychological and organizational scales whereas the term *test* refers to multiple choice tests.

Summary

Even though we have sought to highlight differences between IRT and factor-analytic ME analysis, there is a growing recognition that similar concepts/procedures can be applied to both CFA and IRT ME analysis (Meade & Lautenschlager, 2004; Stark, Chernyshenko, & Drasgow, 2006). While there will still be underlying differences in data assumptions and response modeling, we expect that many

of the aforementioned differences will be reduced as psychometric research advances and researchers broaden the use of IRT and CFA to different settings.

Review of IRT ME/DIF Procedures

In this section, we review how IRT ME procedures have been applied in psychological and organizational research. The goal is to summarize commonly used methods. From this section onward, we use the term *differential item functioning* instead of ME as it is the conventional terminology and the statistics/procedures associated with IRT ME analysis is typically labeled as *DIF* analysis.

To examine how IRT DIF is applied in the field, we conducted a literature search by using key words *IRT* or *item response theory* and *measurement equivalence* or *differential item functioning* or *DIF* in two databases that are most relevant to psychological and organizational research: PsycINFO and EBSCOhost Business Source Complete. No restrictions on the timeframes were set. An article is included in this review only if it satisfies all of the following criteria: (a) It is a peer-reviewed journal article, (b) it includes at least one empirical study on DIF using an IRT approach, and (c) the DIF analysis was conducted on a construct that is frequently studied in organizational research, such as cognitive ability, noncognitive individual difference (e.g., personality, vocational interest, and emotional intelligence), job attitudes, and job performance. There were 51 studies included in the final review. Results of literature reviews are summarized in the following. We note that some studies used multiple samples, measures, or methods, so the total percentage within one category may exceed 100%.

Type of Constructs

The most frequently examined content in DIF literatures are noncognitive individual differences (18 studies; 35.3%), including 9 studies on personality, 4 studies on vocational interests, and 6 studies on other constructs, such as emotional intelligence and self-esteem. Twelve studies (23.5%) examined DIF on attitudes (including 8 studies on job satisfaction). Eleven studies (21.6%) focused on job-related behaviors, such as job performance (5 studies) and sexual harassment (2 studies). There are also 11 studies (21.6%) examining DIF on cognitive ability tests. This demonstrates the wide applicability of DIF measures to different types of constructs that are of interest to organizational researchers as IRT is not only used in the testing context.

Types of Groups Compared (Sources of DIF)

IRT DIF analysis has also been applied across a variety of group dimensions reflecting interest in different potential sources of DIF. Eighteen studies (35.3%) conducted IRT DIF analysis across female and male samples to explore gender-related DIF. Cross-cultural differences are the second most popular DIF factor, with 10 studies (19.6%) examining the DIF of a scale across different nations (e.g., U.S. vs. China) or a scale in different languages (e.g., English vs. Spanish version). Another interesting source of DIF examined is rating source. Six studies (11.8%) investigated DIF between self and other ratings of job performance. Researchers are also interested in how testing conditions influence DIF. Five studies (9.8%) attempted to show DIF (or lack thereof) between a measure under research conditions and the same measure under selection conditions. Other major DIF factors examined include race or ethnicity groups (6 studies; 11.8%), time lags (5 studies; 9.8%), and tenure or age (4 studies; 7.8%).

Sample Size

In IRT analysis, a sample size over 500 has been suggested for good item parameter recovery (Hulin, Lissak, & Drasgow, 1982). Among 43 studies that reported sample sizes, 22 studies (51.2%) used at least one group with fewer than 500 respondents. The average total sample size (combining all groups) across studies is 6,888, which is, however, highly influenced by a few studies with extremely big data sets (e.g., 71,450). The median total sample size across all studies is 1,723. We also computed the difference in sample sizes between the groups compared in each study. Sixteen studies (37.2%) compared groups with equal sample sizes (i.e., difference equals zero). The mean and median differences in sample sizes are 2,716 and 124, respectively. In general, our review shows that studies applying IRT ME procedures have been applied to appropriate sample sizes as suggested in the past.

Test/Scale Length

Ten studies (19.6%) examined DIF on cognitive ability or knowledge tests. The number of items per test ranged from 11 to 210, with a median of 27. The other 41 studies (80.4%) examined noncognitive scales, such as personality, attitudes, and performance. The scale length ranged from 4 to 30, with a median of 10. In general, this shows that IRT can be potentially be applied to a variety of scale lengths.

IRT Models

In Table 4, we describe, not exhaustively, a number of the IRT models that are used in organizational and psychological research. We explicate the different IRT models and the corresponding numbers of responses (dichotomous or polytomous), type of categorical variable (nominal or ordered), type of parameters, and other notable aspects (which include interpretation of parameters and how such models are commonly used). We highlight three main reasons for the multitude of IRT models and when choosing certain models may be appropriate:

1. Number of parameters modeled: For example, while the form of the model is equivalent, the numbers of parameters modeled for the one-parameter logistic model (1PLM), two-parameter logistic model (2PLM), and three-parameter logistic model (3PLM) are different. This can also be seen in the difference between the partial credit model (PCM) and the generalized partial credit model (GPCM). The number of parameters modeled often depends on the type of psychological data. In general, when the data are dichotomous, a 2PLM is usually used for psychological scales (e.g., personality, attitudes, interests) that have a yes/no, agree/disagree, or true/false response format; a 3PLM is used for tests that have a multiple choice response format coded as right/wrong as guessing likely occurs. When there is ambiguity, a comparison of IRT model-data fit should be undertaken to show that one model fits better than another.
2. How categorical responses are treated: For instance, the nominal response model (NRM) would be used when the scale responses are nominal whereas other models assume that scale responses are ordered. Critically, because almost all psychological scales have categorical responses (e.g., disagree, agree), researchers have to make decisions as to whether to treat categorical scale responses as ordered or nominal. Conventionally, we usually treat rating scales (e.g., *strongly disagree* to *strongly agree*) as ordered, and it is our opinion that most rating scales would fare well with ordered IRT models. Researchers interested in more unique applications can use IRT to show that certain categories (e.g., neither agree nor

Table 4. Description of different item response theory (IRT) models.

IRT Model	Number of Responses	Type of Categorical Variable	Parameters	Notes
1 parameter logistic model (PLM)	Dichotomous	Ordered	b_i – item location	b_i – corresponds to trait level where there is 50% probability of responding correctly
2PLM	Dichotomous	Ordered	a_i – item discrimination b_i – item location	b_i – corresponds to trait level where there is 50% probability of responding correctly Typically used for dichotomous yes/no scale items
3PLM	Dichotomous	Ordered	a_i – item discrimination b_i – item location c_i – lower asymptote (guessing)	b_i – corresponds to trait level where there is about > 50% probability (incorporating guessing) of responding correctly Typically used for multiple choice test items where guessing likely occurs
Graded response model (GRM) (Samejima, 1969)	Polytomous	Ordered Items can have different numbers of response categories	a_i – item discrimination b_{ik} – item thresholds For example, thresholds for four response categories: $b_{i1} - 0$ vs. 1,2,3 ($P \geq 1$) $b_{i2} - 0,1$ vs. 2,3 ($P \geq 2$) $b_{i3} - 0,1,2$ vs. 3 ($P \geq 3$)	b_{ik} – corresponds to trait level where there is 50% probability of responding in that category or higher A 2PLM is used to model cumulative response curves for an item as shown in Figure 1C Typically used for Likert-type (rating) scale items
Partial credit model (PCM) (Masters, 1982)	Polytomous	Ordered Items can have different numbers of response categories	b_{ik} – item “steps” (or boundaries/ intersections) For example, steps for 4 response categories: $b_{i1} - 0$ vs. 1 ($P = 1$) $b_{i2} - 1$ vs. 2 ($P = 2$) $b_{i3} - 2$ vs. 3 ($P = 3$)	This is the 1PLM equivalent for polytomous data. A 1PLM is used to model a pair of successive adjacent categories for an item. b_{ik} – corresponds to trait level where responding to the higher category has a higher probability Useful for scoring open-ended test items (e.g., multipart math problem).
Generalized partial credit model (GPCM) (Muraki, 1992)	Polytomous	Ordered Items can have different numbers of response categories	a_i – item discrimination b_{ik} – item “steps” (or boundaries/ intersections) For example, steps for 4 response categories:	Is a generalization of the PCM as it models item discriminations

(continued)

Table 4. (continued)

IRT Model	Number of Responses	Type of Categorical Variable	Parameters	Notes
Rating scale model (RSM) (Andrich, 1978)	Polytomous	Ordered Items must have the same number of response categories	$b_{i1} - 0$ vs. 1 ($P = 1$) $b_{i2} - 1$ vs. 2 ($P = 2$) $b_{i3} - 2$ vs. 3 ($P = 3$) b_i – item location τ_k – category threshold parameters	The PCM is a more general form of the RSM. The RSM is also seen as an extension of the IPLM. The τ_{ik} represents the category threshold relative to the item location.
Generalized graded unfolding model (GGUM) (Roberts Donoghue, & Laughlin, 2000)	Dichotomous/polytomous	Ordered Items can have different numbers of response categories	α_i – item discrimination δ_i – item location τ_{ik} – subjective response threshold parameters	The GGUM is an ideal point model where a low score on an item (e.g., strongly disagree) occurs when $ \theta - \delta_i $ is large. τ_{ik} represents subjective choice of a response category relative to $\theta - \delta_i$. τ_{ik} is symmetric around $\theta - \delta_i$ because a low score (i.e., disagreement) can occur because an individual is too high or too low from the item location.
Nominal response model (NRM) (Bock, 1972)	Polytomous	Nominal Items can have different numbers of response categories	a_{ik} – item option slopes b_{ik} – item option intercepts For example, slopes and intercepts for 4 options: $a_{i0}, b_{i0} - 0$ vs. 1,2,3 ($P = 0$) $a_{i1}, b_{i1} - 1$ vs. 0,2,3 ($P = 1$) $a_{i2}, b_{i2} - 2$ vs. 0,1,3 ($P = 2$) $a_{i3}, b_{i3} - 3$ vs. 0,1,2 ($P = 3$)	Each item parameter corresponds directly to the probability of response for an option. Each item option has its own discrimination and thresholds. A common restriction of $\sum_1^k a_{ik} = 1$ and $\sum_1^k b_{ik} = 1$ imply that the discrimination and threshold parameters are scaled within each item and may not be comparable across items. Typically used for distractor analysis in multiple choice test items.

disagree or neutral) may not necessarily follow the expected ordering (e.g., Hernandez et al., 2004).

3. Assumptions about the response process: Despite common assumptions about ordered scale response, the generalized graded unfolding model (GGUM) assumes that the response function is non-monotonic whereas other models assume monotonicity. That is, the GGUM assumes that the probability of a higher score is larger when the difference between the respondent's standing on the latent trait and the item location is smaller, whereas monotonic models (e.g., graded response model [GRM], PLMs) assume that higher standing on the latent trait results in a higher score. More recently, the GGUM has been advocated for self-reported typical behaviors (i.e., scales), whereas monotonic models are better suited for maximal behaviors (i.e., tests) (Tay & Drasgow, 2009; Tay & Drasgow, 2012b). We note that the current application of monotonic models to self-reported typical behaviors is usually acceptable as most scales have been created under the monotonic assumptions (e.g., classical test theory). However, recent developments show that specific questions such as scale bipolarity can be better addressed using the GGUM (e.g., Tay & Drasgow, 2012b).

Within psychological and organizational research, the most popular IRT models adopted for dichotomous data are the 2PLM and the 3PLM. Sixteen studies (66.7%) used the 2PLM, and 7 studies (29.2%) used the 3PLM. For polytomous data, the GRM (Samejima, 1969) was the most popular. Seventeen studies (60.7%) used GRM out of 28 studies with polytomous data. Studies also included the use of other IRT models such as the PCM (Masters, 1982) and nonparametric models, such as the kernel smoothing method (Ramsay, 1991). We explicate regularly used models including the 2PLM, the 3PLM, and the GRM here and use them to illustrate IRT ME analysis. The 2PLM is shown in Table 1. The 3PLM, where the probability of endorsing (or correctly responding) to item i for a given latent trait level θ_j is given by

$$P(y_i = 1|\theta_j) = c_i + \frac{1 - c_i}{1 + \exp[-1.702a_i(\theta_j - b_i)]},$$

where i and j index the item and the respondent, respectively; the item parameters a_i , b_i , and c_i represent the item discrimination, item location, and item "pseudo-guessing" parameters, respectively.

For polytomous data, the GRM specifies that the probability of endorsing response option (option response curves shown in Figure 1D) k ($k = 0, 1, \dots, m$) for an $m + 1$ point scale is given by:

$$P(y_i = k|\theta_j) = P_{ik}^*(\theta_j) - P_{i(k+1)}^*(\theta_j),$$

where $P_{ik}^*(\theta_j)$ is defined as the probability of endorsing equal to or above response option k using a 2PLM (cumulative response curves shown in Figure 1C) formulated as

$$P_{ik}^*(\theta_j) = P_{ik}^*(y_i \geq k|\theta_j) = [1 + \exp(-1.702a_i(\theta_j - b_{ik}))]^{-1}$$

and

$$P_{i0}^*(\theta_j) = 1$$

and

$$P_{i(m+1)}^*(\theta_j) = 0.$$

The parameters a_i and b_{ik} represent the item discrimination and item thresholds, respectively.

In the illustration section, we focus on procedures that are commonly applied with an emphasis on the 2PLM, the 3PLM, and the GRM as these are the most widely used. While ideal point models

(e.g., the GGUM) represent an exciting area of new research, a detailed discussion of ideal point models is beyond the scope of the current article (for details, see Drasgow, Chernyshenko, & Stark, 2010; Tay & Drasgow, 2012b; Wang, Tay, & Drasgow, 2013).

Assessing Unidimensionality and Model-Data Fit

As shown in Table 2, an initial step to IRT DIF analysis is assessing the assumption of unidimensionality. At this juncture, it is important to note that in unidimensional IRT, the assessment of unidimensionality and model-data fit are distinct but empirically closely related. For example, despite underlying unidimensionality, there may be a misapplication of the IRT model (e.g., 1PLM vs. 3PLM) that could result in poor model-data fit. Therefore, the form of the IRT model is important.

Research has shown that unidimensional IRT models are fairly robust to multidimensionality (Drasgow & Parsons, 1983). It is more important that the data are essentially unidimensional; that is, there is a large general dimension with only small subfactors. As such, examining whether essential unidimensionality holds in the data before conducting DIF analyses is critical because it is possible to obtain DIF due to violations of unidimensionality.

Although essential unidimensionality is important for unidimensional IRT models, only 32 of the reviewed studies (62.7%) explicitly stated the method used to assess unidimensionality. Among those studies, the most frequently used approach is exploratory factor analysis EFA. Twenty-four studies (75%) conducted EFA and examined the eigenvalue of as well as the variances explained by the first extracted factor to determine unidimensionality. Only 4 studies (12.5%) used CFA and 3 studies used parallel analysis (PA) to assess unidimensionality. Our recommendation is that more effort should be used to assess unidimensionality when conducting IRT ME analyses and more generally IRT analyses. We propose some ways to assess unidimensionality and statistics to report in Table 5.

Even though it is necessary to assess whether the IRT model fits the data well after item parameter estimation, only 22 studies (43.1%) clearly mentioned that at least one method was used for assessing model fit. Most of the studies (17 studies, 77.2%) have focused on the use of the adjusted χ^2 statistic (Drasgow, Levine, Tsien, Williams, & Mead, 1995). Six studies (27.3%) also looked at information criteria (e.g., Akaike Information Criterion [AIC], Bayesian Information Criterion [BIC], Consistent Akaike's Information Criterion [CAIC]).

We propose that future research incorporate more tests and reports of model-data fit as shown in Table 5. In particular, we note that the software IRTPRO provides a number of recently developed statistics for assessing unidimensionality and IRT fit. We illustrate the use of these indices later.

Different DIF Indices and Procedures

In this section, we review the different ways DIF has been operationalized in the literature. The review is not exhaustive and is only a sampling of a large number of statistics (see reviews by Penfield & Camilli, 2007; Potenza & Dorans, 1995; Zumbo, 2007). In this section, we present: (1) the concept of the reference and focal group, (2) linking and iterative linking, (3) usage of known anchors, (4) two-step procedure, and (5) the IRT with covariates (IRT-C) procedure.

A summary of DIF indices is presented in Table 6. The Lord's chi-square approach, the differential functioning of items and tests (DFIT) framework, and the likelihood ratio approach are the three major methods used to detect DIF. In particular, 10 studies (19.6%) computed Lord's χ^2 , 21 studies (41.2%) examined used the DFIT framework, and 15 studies (29.4%) adopted the likelihood ratio model comparison approach to detect DIF. Other studies used nonparametric approaches, such as examining item characteristic curves (3 studies; 5.9%), the Mantel-Haenszel approach (2 studies; 3.9%), and the simultaneous item bias (SIBTEST) approach (3 studies; 5.9%). Newer tests for DIF

Table 5. Assessing item response theory (IRT) unidimensionality and model-data fit.

Index/Analysis	Interpretation of Index/Analysis	Strengths/Limitations	Notes/Recommendations	Software
Exploratory factor analysis (EFA)	Examining whether there is a dominant first factor through a visual examination of scree plot; assessing the ratio of the first eigenvalue to the second eigenvalue	-Subjective method -Assumption of continuous scale response does not conform to ordinal IRT models	-Use ordinal EFA for visual inspection and report ordinal EFA fit statistics where possible -Past research recommends that variance accounted for by first factor should be larger than 20% (Reckase, 1979) -Another rule of thumb is that the first eigenvalue is at least 4 to 5 times larger than the second eigenvalue (Lord, 1980)	-SPSS can be used for EFA -Mplus can be used for regular and ordinal EFA
Parallel analysis (PA) (Horn, 1965)	-Comparing observed eigenvalues with bootstrapped eigenvalues of random data generated from the same sized data; when the bootstrapped eigenvalue is larger than the observed eigenvalue, it indicates the associated dimension is mostly random noise -It is expected that the observed second eigenvalue is smaller than the bootstrapped second eigenvalue (average or 95% percentile).	-Is not overly strict and may allow for slight violations of unidimensionality fulfilling the assumption of essential unidimensionality -Assumption of continuous scale response does not conform to ordinal IRT models	-Prior research shows that regular PA to dichotomous data performs well despite linear assumption (Weng & Cheng, 2005) -Regular PA may not be appropriate for test data that has guessing, and modified PA (MPA) may be more appropriate (Drasgow & Lissak, 1983)	-R or SPSS syntax can be found for applying PA -The "ltm" package in R implements MPA (Rizopoulos, 2006)
Confirmatory factor analysis (CFA)	Assessing whether a unidimensional CFA produces reasonable good fit using SEM indices such as Comparative Fit Index (CFI), Tucker-Lewis Index/ non-normed fit index (TLI/ NNFI), root mean square error of approximation (RMSEA), standardized root mean square residual (SRMR)	-Assumption of continuous scale response does not conform to ordinal IRT models	-Report SEM fit statistics in tests of unidimensionality -It is not known whether CFA procedures are overly stringent for tests of essential unidimensionality in IRT	-Mplus can be used for regular and ordinal CFA (e.g., Olino et al., 2012)

(continued)

Table 5. (continued)

Index/Analysis	Interpretation of Index/Analysis	Strengths/Limitations	Notes/Recommendations	Software
Adjusted chi-square statistic (Drasgow, Levine, Tsien, Williams, & Mead, 1995)	-One-way (singles; single items), two-way (doubles; item pairs), and three-way (triples; triplets of items) marginal tables between model-predicted frequencies and observed frequencies are used to obtain a chi-square value. A higher chi-square value would indicate worse fit. The chi-square value is adjusted to a fixed sample size of 3,000. -It is proposed that singles, doubles, and triples with values < 3 indicate good model-data fit.	-Simple heuristic -IRT-based metric	-New research shows that only doubles and triples are sensitive to misfit. Further, the absolute value of < 3 is a poor indicator the value varies by sample size and scale length (Tay & Drasgow, 2012a) -Recommended that the doubles and triples adjusted chi-square be used primarily for assessing relative fit.	-MODFIT software (Stark, 2001)
M ₂ statistic (Maydeu-Olivares & Joe, 2005, 2006)	-This statistic is based on one-way and two-way marginal tables to test the null hypothesis that the observed respondent response data is not significantly different from the modeled data response. -For good model-fit, we expect that the M₂ statistic is not significant ($p > .05$). Further, the RMSEA computed using the M₂ value should be close to zero.	-IRT-based metric -Provides a simple and parsimonious way of assessing overall model-data fit	-Requires more research to determine when associated RMSEA value and p value indicate good fit.	IRTPRO (Cai, Thissen, & du Toit, 2011)
Standardized local dependence (LD) χ^2 (Chen & Thissen, 1997)	-This statistic reveals the extent a pair of items is associated beyond the underlying posited latent dimension. Specifically, the standardized LD-χ^2 is computed based on an approximate χ^2 for observed and modeled frequencies for each item pair and then standardized (so that items with different numbers of response categories can be compared).	-IRT-based metric -Allows the identification of items that are correlated beyond the underlying factor (local dependency)	-Report the standardized LD-χ^2 when using IRTPRO	IRTPRO

(continued)

Table 5. (continued)

Index/Analysis	Interpretation of Index/Analysis	Strengths/Limitations	Notes/Recommendations	Software
	-It is suggested that values larger than 3 indicate LD that suggests of multidimensionality. Examining the matrix of standardized LD-χ^2 can reveal which set of items may be locally dependent and require fitting an additional dimension.			
S- χ^2 statistic (Orlando & Thissen, 2000, 2003)	-This statistic tests the null hypothesis that the observed responses across categories for a single item are not significantly different from the modeled responses. -For good model-fit, we expect that most items would exhibit nonsignificant p values ($p > .05$).	-IRT-based metric -Allows the assessment of fit for specific items	-Report the S- χ^2 statistic when using IRTPRO	IRTPRO
Bivariate residual (BVR)	Similar concept to standardized LD χ^2. Each pair of items has a BVR. Larger values (>3) for item pairs indicate association beyond latent dimension.	-IRT-based metric when an IRT response model is applied -Allows the identification of items that are correlated beyond the underlying factor (local dependency)	-Report the average BVR value when using Latent GOLD	Latent GOLD (Vermunt & Magidson, 2013)
Information criteria	-Assesses the relative parsimony of IRT models; lower values of information criteria indicate better fit -Indicators include: Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Consistent Akaike's Information Criterion (CAIC),	-Cannot be used to assess absolute model-data fit -Useful for relative fit	-Report one of the information criteria when comparing fit between multiple IRT models	IRTPRO Latent GOLD BILOG (Zimowski, Muraki, Mislevy, & Bock, 1996) MULTILOG (Thissen, Chen, & Bock, 2003)
Ratio between log-likelihoods (Thissen, Steinberg, & Wainer, 1993)	-Compares models to determine if IRT models with DIF are significantly different than those without (or less) DIF	-Cannot be used to assess absolute model-data fit		IRTPRO Latent GOLD BILOG MULTILOG

Table 6. Different item response theory (IRT) differential item functioning (DIF) indices and associated procedure(s).

DIF Test	Description	Associated Procedure(s)	Software
Lord's χ^2 (Lord, 1980)	Focus on estimated item parameter difference. Tests if there is a significant difference between estimated item parameters (e.g., a_i and b_i parameters for a 2 parameter logistic model [PLM]) in the focal group as opposed to the reference group. Initial application for dichotomous IRT models. A generalization was developed to assess DIF in multiple groups (Kim, Cohen, & Park, 1995).	Linking Iterative linking	-ITERLINK (Stark, 2002)
Wald χ^2	Focus on estimated item parameter difference. This is an improvement over Lord's χ^2. Tests if there is a significant difference between estimated item parameters in the focal group as opposed to the reference group. Improved estimation of asymptotic sampling variance-covariance matrix (e.g., standard errors) results in better controlled Type I errors (Langer, 2008; Woods, Cai, & Wang, 2012). Appropriate standard errors are also computed in the software Latent GOLD (Vermunt & Magidson, 2008, 2013). Can be applied to dichotomous and polytomous IRT models. Can be applied to multiple groups.	Use known anchors to test other items for DIF Use 2-step procedure to test all items for DIF	IRTPRO (Cai, Thissen, & du Toit, 2011) Latent GOLD (Vermunt & Magidson, 2013)
Differential item functioning of items and tests (DFIT) (Raju, van der Linden, & Fleer, 1995)	Focus on estimated item characteristic curve differences. Tests if there is a significant difference between the reference group item characteristic curve (ICC) and the focal group item characteristic curve. Specifically, the squared difference between the ICCs is weighted by the focal group trait density function. Can be applied to dichotomous and polytomous IRT models.	Linking Iterative linking	DFIT (Raju et al., 1995) ITERLINK

(continued)

Table 6. (continued)

DIF Test	Description	Associated Procedure(s)	Software
Likelihood ratio (Thissen, Steinberg, & Gerrard, 1986; Thissen, Steinberg, & Wainer, 1993)	Can also examine differential test functioning. Focus on the likelihood ratios between two nested IRT models. Tests whether the less constrained model (which corresponds to the instance where there is DIF on a single item or multiple items) is significantly different from the more constrained model (which corresponds to the instance where there is no DIF). Can be applied to dichotomous to polytomous IRT models. Can be applied to multiple groups.	Use known anchors to test other items for DIF	BILOG (Zimowski Muraki, Mislevy, & Bock, 1996) MULTILOG (Thissen, Chen, & Bock, 2003) IRTPRO Latent GOLD
Mantel Haenszel (MH) (Holland & Thayer, 1988)	Focus on comparing proportion of correct responses in the reference and focal groups. Both groups are divided into k trait levels using total observed score. At each level, a 2 (group: reference or focal) $\times$ 2 (response: 1 or 0) contingency table is used for calculating a common odds ratio across all k trait levels (as a measure of DIF effect size) and for calculating an overall χ^2 statistic. Initially developed for dichotomous items and comparison between two groups. Extensions allow testing of polytomous items and across multiple groups (Fidalgo & Madeira, 2008; Penfield, 2001; Zwick & Thayer, 1996).	—	DIFAS (Penfield, 2005) EASY-DIF (Gonzalez, Padilla, Hidalgo, Gomez-Benito, & Benitez, 2011)
Simultaneous Item Bias (SIBTEST) (Shealy & Stout, 1993)	Focus on comparing proportion of correct responses in the reference and focal groups. The test is divided into a valid subtest and a potential DIF subtest. Using the valid subtest, score groups are created for the reference and focal groups. For each valid subtest score group, the difference between reference and focal group potential DIF subtest total score is calculated; this is aggregated and adjusted	Use known anchors to test other items for DIF	DIF-Pack http://psychometrictools.measuredprogress.org/

(continued)

Table 6. (continued)

DIF Test	Description	Associated Procedure(s)	Software
	for potential trait mean difference between groups. Initially developed for dichotomous IRT models. Requires known non-DIF (anchor) items. A SIBTEST statistic is produced for all potential DIF items (not individual items). Extensions allow testing of polytomous items (Chang, Mazzeo, & Roussos, 1996)		
Bivariate residual (BVR) and Wald χ^2 (Vermunt & Magidson, 2005)	Focus on flagging potential DIF items and testing them. Using a generalized latent variable modeling framework, the BVR is a Pearson χ^2-type measure reflecting the discrepancy between observed and expected cell counts in the two-way cross-tabulation between a covariate (or an observed grouping variable) and an item. In a subsequent model, the associated item parameter is tested with the Wald χ^2 statistic. Useful for ascertaining DIF over multiple group characteristics (e.g., age and gender simultaneously).	IRT-C	Latent GOLD

have been developed using a general latent variable modeling approach (i.e., IRT-C), which uses the Wald χ^2 test to test for significance and the bivariate residual (BVR) to identify potential DIF. In the following, we further discuss the related concepts and procedures that are used with these DIF statistics.

Reference Group, Focal Groups, and Covariates. Typically, IRT DIF analysis is conducted between two groups. In the testing context, the reference group is often the majority sociodemographic group that has a larger sample; the focal group is the minority sociodemographic group with a smaller sample. Designation of the reference or focal group is arbitrary and does not affect the computation of DIF. In multigroup comparisons, there may be three or more groups implying one reference group and multiple focal groups. With newer DIF procedures, such as IRT-C, multiple group characteristics (i.e., covariates; e.g., age, race, gender) can be considered simultaneously. Analogous to multiple regression, the reference “group” corresponds to when group characteristic variables are coded as zero. For example, gender (0 = male; 1 = female) and age (grand-mean centered) would result in a reference “group” of average-aged males in the sample.

Linking and Iterative Linking. With IRT, item parameters are independent of the samples of participants/examinees from which they are estimated. Item parameter sample independence however does not imply that estimated parameters are identical (save random error) when different samples are used. This is because in the IRT estimation, the latent trait is typically scaled to be $N(0,1)$ to resolve scale indeterminacy. Thus, item parameters estimated in two different samples are interchangeable only after linear transformations are applied. More concretely, using a sample of high-ability test takers, estimated item parameters will appear to be easy and less discriminating. Using a sample of test takers that span a wide ability range, estimated item parameters will appear to be moderate and discriminating. The difference in item parameters is due to the arbitrary latent trait scaling. This procedure of linearly transforming the item parameters so they are on the same scale across samples is called linking, or equating. We note that linking is unnecessary if the latent trait distributions are known; that is, we have knowledge about the trait distribution of the focal group relative to the referent group $N(0,1)$ distribution.

For some DIF indices, such as the Lord's χ^2 or DFIT, separate IRT item parameter estimations are undertaken for the reference group and the focal group, which are then linked to be on the same metric so that differences in item parameters or item response curves can be calculated. In the presence of DIF items however, the linking procedure is less accurate because these items are not invariant across groups. As such, an iterative linking approach is proposed so that linking is based on only non-DIF items (Candell & Drasgow, 1988): (a) Item parameters are separately estimated in each sample then linked and placed on a common metric, (b) the DIF test is then applied to determine whether linked item parameters are significantly different across the samples, (c) items that are not significantly different are used to relink the item parameters, and (d) the process can stop (for two-stage linking) or Steps b and c are repeated iteratively until the same set of DIF items are confirmed (for iterative linking).

Using Known Anchors. A common approach to assessing DIF is a known-anchor approach where items that are known to be measurement invariant across groups are used to put all items common metric. This process is implemented in a simultaneous estimation of both reference and focal group item parameters when using the Wald χ^2 and the likelihood ratio test. For the Wald χ^2 , each potential DIF item can be tested for significant difference between the reference and focal group. For the likelihood ratio test, the less constrained model (M_{LC}) freely estimates potential DIF items; the more constrained model constrains potential DIF items (M_C). Across both models, anchor items are constrained to be equal. The -2 log-likelihood ratio of M_{LC}/M_C , which is χ^2 -distributed with degrees-of-freedom (df) equal to df difference between M_C and M_{LC} , can be tested for significant difference. SIBTEST uses known anchors to calculate a DIF statistic for all other items. A limitation of the known anchor approach is that it is seldom known which items are invariant prior to DIF testing.

Two-Step Procedure. The two-step procedure of testing for DIF (Woods, Cai, & Wang, 2012) using the Wald χ^2 can be implemented in IRTPRO to overcome the need for known anchors as these are typically not known a priori. The two steps include: (a) constraining all items to be equal across group(s) and estimating the latent trait distributions (mean and variance) for the focal group(s); (b) focal group(s) trait distributions are fixed at values found in Step a and item parameters are freely estimated across groups. All items in this second step can be tested for DIF as they are scaled on a common metric (i.e., latent trait distributions are known and we do not need to specify linking/anchor items). Despite high power to detect DIF, a potential problem with the two-step procedure is that the Type I error rate is much higher than the nominal rate when the proportion of DIF items is substantial (Woods et al., 2012). We propose that the best way to detect DIF when anchor items are not known a priori is to use the two-step procedure for identifying anchor items because of the

high power to detect DIF. Using the anchor items found from the two-step procedure, an iterative procedure can be used to test for DIF in all other non-anchor items (as the Type I errors are well controlled when anchor items are known). This will be demonstrated in our illustration section.

IRT-C Procedure. The IRT with covariate approach is based on generalized latent variable modeling. The bivariate residual is an index that is used to flag for potential DIF as it indexes the additional association between the item and the covariate (i.e., group characteristic) not accounted for by the IRT model specifying invariance. The IRT-C procedure for identifying DIF items is iterative (Tay et al., 2013): (a) Identify potential DIF items by assessing item-covariate BVR; (b) test DIF for the largest item-covariate BVR using the Wald statistic; (c) if Step b Wald statistic is significant, allow item to be freely estimated for the item-covariate and repeat Steps a and b to test other item-covariates; if Step b Wald χ^2 statistic is not significant, end procedure.

Some issues to note are that certain popular procedures such as the Mantel Haenszel (MH) and the SIBTEST are based on observed scores and not a latent trait model (i.e., nonparametric approaches). Researchers interested in using a latent trait approach should consider the other DIF tests and associated procedures. Another issue is that there is often no pre-knowledge of anchor items in the subpopulations of interest. As such, procedures that do not require known anchors are preferred. Finally, there is also the issue of convenience. DIF tests that do not require multiple software packages allow a more seamless DIF analysis. As such, the DIF indices that are recommended include: the Wald χ^2 , the likelihood ratio test, and the BVR/Wald χ^2 . In our illustration, we focus on the use of the Wald χ^2 with IRTPRO and the BVR/Wald χ^2 with Latent GOLD. The former index/procedure is useful for a variety of two-group and multigroup DIF analyses; the latter index/procedure is unique as it can be used to assess DIF on multiple group characteristics. This is similar to the newest developments in CFA ME research that uses multiple indicators multiple causes (MIMIC) models (Woods, 2009; Woods et al., 2012; Woods & Grimm, 2011; Woods, Oltmanns, & Turkheimer, 2009).

Effect Size Estimates

The previous section discusses some of the more common significance testing procedures for DIF. However, as with other significance tests (e.g., t tests, correlations, etc.), the power of DIF tests is affected by sample size, and small differences can reach statistical significance simply because the analysis is run with large samples (which are often recommended with IRT analyses). Estimating effect sizes for DIF and DTF is important as a statistically significant result does not necessarily imply a practically meaningful difference. As with any statistical testing, it is important to examine statistical significance but also effect sizes and by extension practical importance of significant DIF (Meade, 2010; Stark et al., 2004).

Recent work has attempted to provide measures of effect size for the amount of DIF/DTF present between groups. There are multiple ways to conceptualize and report DIF effect sizes (DeMars, 2011; Meade, 2010; Steinberg & Thissen, 2006). We highlight two approaches here: (a) a rigorous expected score effect size approach (sophisticated) and (b) a heuristic latent trait estimation approach (simple).

Expected Score Approach. With regard to a rigorous expected score effect size approach, the most common effect size index used in organizational research is a variation of expected scores. For a given trait level, one can obtain an expected item score given the set of estimated item parameters. By extension, one can obtain an expected test score for a trait level given all the estimated item parameters. Expected scores have the advantage of being intuitively interpretable in the metric of the original measures. Expected scores can be computed for both dichotomous test items (where the expected score for each item is just the P value for that item for the level of theta) or polytomous

items. The expected score for polytomous item i can be computed as the sum of the probabilities of a response to each of the m response options times the observed score value of that option (X_k):

$$ES_{s(\hat{\theta})i} = \sum_{k=1}^m P_{ik}(\hat{\theta}) X_{ik}, \quad (1)$$

where $ES_{s(\hat{\theta})i}$ is the expected score for respondent s (with a given estimated theta value) for item i , $P_{ik}(\hat{\theta})$ is the probability of responding to category k given the respondent's estimated theta level, and X_{ik} is the value of response option k (e.g., X_{im} would be 5 for an item with 5 response options). The expected score is similar to an item-level true score and has a potential range from the lowest response option (e.g., 1) to the highest (e.g., 5).

For both polytomous and dichotomous items, expected scale (or test) scores (ETS) for each respondent s can be computed as the sum of the expected item scores for person s :

$$ETS_s = \sum_{i=1}^j ES_{si} \quad (2)$$

The use of expected scores as a measure of effect size in organizational research is implicit in the DFIT approach in which NCDIF (noncompensatory DIF/uniform DIF) is a direct measure of the difference in expected scores across groups. However, Raju's (1988) work tended to emphasize significance testing with NCDIF more than interpretation of NCDIF as an effect size. Stark et al. (2004) expanded upon Raju, van der Linden, and Fleer's (1995) work to derive a series of test/scale-level effect size measures. Meade (2010) developed a taxonomy of potential effect size measures that could be computed based on both item-level and test-level scores. This taxonomy incorporated NCDIF as well as Stark et al.'s indices. Meade recommended reporting signed item difference in the sample (SIDS), which is simply the average, across focal group respondents, of the difference in expected scores computed using the focal group's item parameters and the reference group's item parameters:

$$SIDS_i = \frac{\sum_{s=1}^N [ES_{(si|\hat{\theta}, \gamma_F)} - ES_{(si|\hat{\theta}, \gamma_R)}]}{N}. \quad (3)$$

The SIDS can be interpreted in the metric of the scale item. Meade (2010) also recommended using a standardized version of differences in expected scores similar to Cohen's d (expected score standardized difference, ESSD):

$$ESSD_i = \frac{\overline{ES}_{(\gamma_F)} - \overline{ES}_{(\gamma_R)}}{SD_{ItemPooled}}. \quad (5)$$

Other item-level indices and their test-level counterparts can be found in Meade (2010).

These indices can be computed manually or using the VisualDF program (see Meade, 2010).

Latent Trait Estimation Approach. In this approach, the latent trait difference between reference and focal group(s) when DIF is not accounted for versus when it is accounted for is used to compute an effect size. This approach is simple and useful when the goal is to determine whether the presence of DIF affects the interpretation of whether reference and focal group(s) have substantially different latent trait levels.

In the simultaneous estimation of latent trait distributions where the reference group latent trait is $N(0,1)$, the estimated latent trait level of the focal group(s) is standardized to the reference group metric. With simultaneous estimation where DIF is not accounted for, the estimated focal group

latent trait level comprises both true latent trait difference (i.e., impact) and bias introduced by DIF. Formulaically, $\hat{\theta}_{focal} = \widehat{Impact} + \widehat{BIAS}$. We term this model *Model TB*, with TB representing True latent trait differences and Bias. With simultaneous estimation where all DIF items are accounted for, the estimated focal group latent trait level comprises only true latent trait difference (i.e., impact). Formulaically, $\hat{\theta}_{focal} = \widehat{Impact}$. We label this model *Model T*, where T represents True latent trait difference.

Using Models TB and T, we can obtain a standardized measure of bias by comparing the latent trait differences estimated in the two models. For example, if Model TB produces a focal group θ_{mean} of $-.30$ and Model T produces a focal group $\theta_{mean} = -.25$, the magnitude of the effect size of the bias is $.05$, which is fairly small. Bias calculated in this manner would approximate Cohen's d , and one can report the extent bias is small ($\sim .2$), medium ($\sim .5$), or large ($\sim .8$ or larger).

Another advantage in this approach is that it allows researchers to determine whether bias from DIF directly affects the results of statistical testing. Specifically, one can determine whether statistical testing of the latent trait difference between reference and focal group(s) is affected by DIF. To continue with the previous example, if Model TB produces a focal group θ_{mean} of $-.30$ ($SE = .05$) and Model T produces a focal group $\theta_{mean} = -.25$ ($SE = .05$), the effect size of the bias is $.05$, which is fairly small. Further, the focal group trait level continues to be significantly lower than the reference group as the SE is substantially smaller than the estimated focal group mean, implying that the observed DIF in items do not change results of statistical testing.

Illustrations

In this section, we present the use of two different programs for assessing IRT ME. We recommend the use of the software IRTPRO (Cai, Thissen, & du Toit, 2011) for basic IRT analysis and two or multiple-group IRT ME analysis. As shown in Table 4, IRTPRO can be used to estimate IRT item parameters and also provides key indices and tests for unidimensionality and goodness of fit. In addition, IRTPRO implements the Wald χ^2 test for assessing IRT DIF. The two-step procedure can be used to identify potential anchor items, and there is no need for a priori knowledge of invariant items. From experience and past research, the Wald χ^2 has well-controlled Type I errors when anchor items are specified (Woods et al., 2012). Finally, IRTPRO also produces the estimated latent trait difference as an index of effect size. The first two illustrations demonstrate how to assess DIF in a two-group and three-group setting.

We also provide an illustration of Latent GOLD (Vermunt & Magidson, 2013) because this implements the latest method of DIF detection, namely, the IRT-C procedure, which allows us to evaluate DIF over multiple group characteristics (i.e., covariates). This is analogous to recent research and interest in MIMIC models for ME analysis in the CFA framework (Woods, 2009; Woods et al., 2009; Woods & Grimm, 2011). We show how to assess DIF using two different group characteristics (i.e., covariates) that can be generalized to a larger number of group characteristics. This procedure is useful when one attempts to determine whether DIF is attributable to a particular group characteristic while controlling for other variables (Tay, Newman, & Vermunt, 2011). For instance, is DIF attributable to Black-White racial difference or to socioeconomic status? Research has shown that the IRT-C procedure has good control of Type I error rates (Tay et al., 2013).

In all three illustrations, we use simulated data so that all instances of DIF and latent trait differences between reference and focal groups are known. In addition, we provide the simulated data, output, parameterization information, and additional instructions for researchers on the website: www.louistay.com. An outline of the procedures for quick reference is shown in Table 7.

Table 7. Recommended item response theory (IRT) steps for differential item functioning (DIF) analysis using IRTPRO/Latent GOLD.

Step	Key Steps and Goals	Procedure/Indicators Used
1	<i>Assess model-data fit and unidimensionality.</i> Seeking to establish the appropriateness of IRT model parameterization and dimensionality for the data at hand	Overall absolute model fit across all groups M_2 statistic (Maydeu-Olivares & Joe, 2005, 2006): For good model-fit, we expect that the M_2 statistic is not significant ($p > .05$). Further, the root mean square error of approximation (RMSEA) computed using the M_2 value should be close to zero. Model fit within each group $S-\chi^2$ statistic (Orlando & Thissen, 2000, 2003): For good model-fit, we expect that most items would exhibit nonsignificant p values ($p > .05$) Standardized local dependence (LD) χ^2 (Chen & Thissen, 1997): Examining the matrix of standardized LD-χ^2 can reveal which set of items may be locally dependent and require fitting an additional dimension. It is suggested that values smaller than 3 indicate good fit. Relative fit Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC): Information criteria statistics can be used to compare models of interest (different model parameterizations [e.g., 1 parameter logistic (PL) vs. 2PL] or models with different dimensionality). IRTPRO DIF statistic Wald χ^2 statistic with accurate error variance-covariance matrix <i>Two group analysis:</i> Significant Wald χ^2 statistic implies significant difference in item parameters between two groups. <i>Multigroup analysis:</i> Wald χ^2 statistic is associated with a specific contrast that tests specific group differences. For example, in a three-group analysis, there are two orthogonal contrasts: (A) 2, -1, -1; and (B) 0, 1, -1. Contrast A tests whether item parameters of Group 1 are significantly different from Groups 2 and 3. Contrast B tests whether item parameters of Group 2 are significantly different from Group 3. LG DIF Statistic Item-covariate bivariate residual (BVR): Wald χ^2 statistic with appropriate standard errors Iterative Procedure In IRTPRO: (1) Identity potential anchor items via the option "test all items, anchor all items." This relies on a two-step procedure in which an initial model with all items constrained as equal across groups is estimated to obtain reference group(s) theta distribution parameters. Then, these theta parameters are fixed in a subsequent model and all items are freely estimated and tested for DIF. Because this has good power but high Type I errors, we use it to identify non-DIF items as anchors.
2	<i>Iterative IRT DIF analysis.</i> Ascertaining DIF without preknowledge of anchor items	

(continued)

Table 7. (continued)

Step	Key Steps and Goals	Procedure/Indicators Used
		(2) Use non-DIF items as anchors to test DIF on all other items. This is the “test candidate items, estimate group difference with anchor items” option. This has high power but well-controlled Type I error rates. (3) If there are tested items that are not significant, add them to the list of anchors and repeat Step 2; if all tested items are significant, end procedure. In LG: (1) Identify potential DIF items by assessing item-covariate BVR; (2) test DIF for the largest item-covariate BVR using the Wald χ^2 statistic; (3) if step 2 Wald χ^2 statistic is significant, allow item to be freely estimated for the item-covariate and repeat steps 1 and 2 to test other item-covariates; if step 2 Wald statistic is not significant, end procedure.
3	<i>Estimate latent group mean difference.</i> Determining the magnitude and significance of group mean difference(s) after establishing anchor items	Difference between group means are estimated after all DIF items are freely estimated across groups
4	<i>Estimating DIF effect size</i>	Heuristic approach Estimate the fully constrained model which has <u>T</u> true mean level differences and <u>B</u>ias (Model TB) Estimate the model with all DIF items modeled which has <u>T</u> true mean level differences only (Model T) Examine the extent Model TB and Model T differ in terms of estimated latent trait differences between reference and focal group(s) (i.e., Model TB θ - Model T θ) Examine whether statistical inferences of latent mean differences differ between Model TB and Model T.

Illustration 1: Two Group 2PLM IRT DIF Analysis

In this first illustration, we demonstrate how to examine DIF between two groups with a dichotomous response scale for a 2PLM. The same procedures for a two-group comparison can be applied to other dichotomous (1PLM, 3PLM) and polytomous (GRM) models.

Data Generation. Using realistic item parameters from Stark, Chernyshenko, and Drasgow (2006), we generated data of 1,000 participants per group for a scale length of 15 items. Using the conversions shown in Table 1, factor-analytic item parameters λ_i and γ_i from Stark, Chernyshenko, and Drasgow’s (2006) study representing item loadings and thresholds, respectively, were converted to normal ogive IRT item parameters $a_i = \frac{\lambda_i}{\sqrt{1-\lambda_i^2}}$ and $b_i = \frac{\gamma_i}{\lambda_i}$, respectively. In the logistic parameterization, a scaling factor of 1.702 is also included as shown in the following so that the same a_i and b_i parameters give rise to the same IRT curve:

Table 8. Illustration 1: simulated item and theta parameters.

	Group = 1 ($\theta_{\text{mean}} = 0, \theta_{SD} = 1$)				Group = 2 ($\theta_{\text{mean}} = 0.2, \theta_{SD} = 1$)				Type of differential item functioning (DIF)
	λ	γ	a	b	λ	γ	a	b	
1	0.90	-0.26	2.06	-0.29	0.90	-0.26	2.06	-0.29	Large ab DIF
2	0.66	-0.06	0.88	-0.09	0.66	-0.06	0.88	-0.09	
3	0.83	-0.42	1.49	-0.51	0.43	0.08	0.48	0.19	
4	0.71	-0.14	1.01	-0.20	0.71	-0.14	1.01	-0.20	
5	0.77	-0.37	1.21	-0.48	0.77	-0.37	1.21	-0.48	Large ab DIF
6	0.68	-0.34	0.93	-0.50	0.68	-0.34	0.93	-0.50	
7	0.58	-0.48	0.71	-0.83	0.18	0.02	0.18	0.11	
8	0.8	-0.07	1.33	-0.09	0.80	-0.07	1.33	-0.09	
9	0.85	-0.30	1.61	-0.35	0.85	-0.30	1.61	-0.35	Large ab DIF
10	0.85	-0.48	1.61	-0.56	0.85	-0.48	1.61	-0.56	
11	0.82	-0.27	1.43	-0.33	0.42	0.23	0.46	0.55	
12	0.8	-0.26	1.33	-0.33	0.80	-0.26	1.33	-0.33	
13	0.85	-0.03	1.61	-0.04	0.85	-0.03	1.61	-0.04	Large ab DIF
14	0.84	-0.14	1.55	-0.17	0.84	-0.14	1.55	-0.17	
15	0.86	-0.27	1.69	-0.31	0.46	0.23	0.52	0.50	

$$P(u_i = 1|\theta_j) = \frac{1}{1 + \exp[-1.702a_i(\theta_j - b_i)]}.$$

Large item DIF on both *a* and *b* parameters was simulated on items 3, 7, 11, and 15, where items were less discriminating (lower *a* value) and more difficult to endorse (larger positive *b* value). Group mean differences were simulated so that the focal group ($\theta_{\text{mean}} = 0.20, \theta_{SD} = 1.0$) had on average higher theta values than the reference group ($\theta_{\text{mean}} = 0.00, \theta_{SD} = 1.0$). The simulated item parameters are shown in Table 8.

Fitting a Unidimensional 2PLM. Table 7 shows the indices for assessing model-data fit and how to interpret the results. Following Table 7, we fit a unidimensional 2PLM to determine whether a unidimensional 2PLM fits the data. We found that there was good overall model-data fit. The M_2 value of 153.27 ($df = 178$) was not significant ($p > .05$) and yielded a low RMSEA value of .00. Second, in each of the groups, the $S\text{-}\chi^2$ item-level fit statistics were generally good. There was only one item (Item 7) for Group 2 that showed significant difference between the observed and modeled response frequencies ($p < .05$). This may be a result of the simulated DIF. Third, we found little evidence for multidimensionality given small standardized local dependence $\chi^2(\text{LD-}\chi^2)$ in both groups where the largest value was 1.00. In general, it is recommended that values smaller than 3 indicate evidence of unidimensionality (Cai et al., 2011). Therefore, we conclude that a unidimensional 2PLM is appropriate for the data and we can test for DIF.

DIF Analysis. Following the procedures in Table 7, we initially fit a fully invariant model as displayed in Table 9 (Model 1). This model assumes that all items are equivalent across groups. Based on the assumption that all the items are invariant, there was no significant difference in theta values across both groups. Nevertheless, the overall probability of the model was low ($M2\ p\ \text{value} = .0001$), and the RMSEA value of 0.04 while reasonably good was poorer than the subsequent fitted models.

Next, we sought to identify anchor items in Model 2. To do so, we used the two-step procedure to test all items for DIF as described in Table 7. Specifically, IRTPRO fixes the latent trait values for

Table 9. Illustration 1: two group differential item functioning (DIF) analysis for 2 parameter logistic model (PLMO).

Model	M2 prob	M2 Square Error of Approximation (RMSEA)	Akaike Information Criterion (AIC)	Bayesian Information Criterion (BIC)	Items With Significant DIF	Anchor Items	Theta Mean (SE)	Comment
1	0.0001	0.04	3,1285.05	3,1464.28	—	—	G1: .00 (—) G2: -.08 (.05)	Full measurement equivalence across items
2	0.9495	0.00	3,0621.02	3,0957.08	2,3*,7*,8,11*,12,15*	—	G1: .00 (—) G2: -.08 (—)	Two step procedure used to test for DIF. Step 1: Model 1 is estimated to obtain Group 2 theta mean and SD; Step 2: Mean and SD values of theta values for Group 2 is used—all items are freely estimated across groups and tested for DIF
3	0.4017	0.00	3,0615.17	3,0872.82	3*,7*,11*,15*	1 [†] ,4 [†] ,5 [†] ,6 [†] ,9 [†] ,10 [†] ,13 [†] ,14 [†]	G1: .00 (—) G2: .13 (.05)	DIF analysis conducted by using all non-DIF items in Model 2 as anchor items
4	0.2138	0.01	3,0609.56	3,0833.60	3*,7*,11*,15*	1 [†] ,2 [†] ,4 [†] ,5 [†] ,6 [†] ,8 [†] ,9 [†] ,10 [†] ,12 [†] ,13 [†] ,14 [†]	G1: .00 (—) G2: .14 (.05)	DIF analysis conducted by using all non-DIF items in Model 3 as anchor items

Note. * represents items that are correctly identified as having DIF; [†] represents items that are correctly identified as having no DIF; G1 and G2 represent the reference group and focal group, respectively.

both groups to that estimated in a fully invariant model (Model 1) and tests all items for DIF as they are all scaled on a common metric. At this step, we found that there were seven items with significant DIF, but the remaining eight items did not exhibit DIF.

From the two-step procedure, we found eight non-DIF items. We used these eight non-DIF items as anchor items in Model 3 to test for DIF in the remaining items using the Wald χ^2 statistic. In Model 3, we found that four out of seven of the items showed DIF (and three items did not show DIF).

We repeated the process with Model 4 where we used all the 11 items that did not exhibit significant DIF in Model 3 as anchor items. Again, the same set of items revealed DIF and we ended the DIF procedure. The final model revealed good overall model fit in that the M2 probability was not significant ($p = .21$) and there was a low RMSEA value of .01. Further, the AIC and BIC values were low indicating good relative model-data fit. With regard to the latent trait mean differences between the reference and focal group, we found an estimated mean difference of .14 ($p < .05$) such that the focal group had a higher theta value. Overall, these results were in line with the simulated data.

Effect Size Estimation. As displayed in Table 9, using the latent trait approach, we found that the difference between the Model TB (Model 1), where DIF was not modeled (accounted for), and Model T (Model 4), where DIF was accounted for, was substantial. This is because the focal group mean was $-.08$ for Model TB and it was estimated at $.14$ for Model T. Given that Model T estimates the “true” latent trait difference without bias, the magnitude of the bias from differential functioning was approximately $|-.08 - .14| = .22$. This represents a small bias effect. Nevertheless, the bias introduced by the items for this particular sample would lead to an incorrect statistical inference that there is no significant mean difference between groups when there is. This can be seen in Model 1 when DIF was not accounted for. When DIF was modeled in Model 4, we see that a significant mean difference could be found between groups as the value of $.14$ was almost three times the size of the standard error of $.05$. As such, the effect of DIF for statistical inference (and subsequent substantive inference) is not negligible.

Illustration 2: Three Group GRM IRT DIF Analysis

In the previous illustration, we presented procedures for a two group IRT DIF analysis. In this illustration, we show how to conduct DIF analysis for three groups through the use of DIF contrasts. This procedure can be generalized to more than three groups. We apply this to polytomous data using a GRM and note that the same procedures can be used for dichotomous data (1PLM, 2PLM, and 3PLM). This procedure is outlined in Table 7.

Data Generation. Using realistic item parameters from Stark, Chernyshenko, and Drasgow (2006) as shown in Table 10, we generated data of 500 participants for each of the three groups for a scale length of 15 items. We varied the degree and type of DIF as shown in Table 4. Group mean differences were simulated so that Groups 1 and 2 did not differ but Group 3 had a lower mean value ($\theta_{\text{mean}} = -.30$). The simulated parameters are shown in Table 10.

Fitting a Unidimensional GRM. As shown in Table 7 we fit a unidimensional GRM to the data to determine the model-data fit and assess unidimensionality. Foremost, we found that there was reasonable overall model-data fit. The M_2 value of 4,840.30 ($df = 4,991$) was significant ($p < .001$) but yielded a low RMSEA value of .00. Given that we fit the appropriate model to the known generating model, this suggests that the M_2 significance testing may be less trustworthy when there is DIF.

Table 10. Illustration 2: simulated item and theta parameters.

	Group 1 ($\theta_{\text{mean}} = 0; \theta_{SD} = 1$)					Group 2 ($\theta_{\text{mean}} = 0; \theta_{SD} = 1$)					Group 3 ($\theta_{\text{mean}} = -.30; \theta_{SD} = 1$)						
	a	b ₁	b ₂	b ₃	b ₄	a	b ₁	b ₂	b ₃	b ₄	Type of DIF	a	b ₁	b ₂	b ₃	b ₄	Type of DIF
1	2.06	-1.34	-0.63	-0.29	0.47	2.06	-1.34	-0.63	-0.29	0.47		2.06	-1.34	-0.63	-0.29	0.47	
2	0.88	-2.15	-0.76	-0.09	1.68	0.88	-2.15	-0.76	-0.09	1.68		0.88	-2.15	-0.76	-0.09	1.68	
3	1.49	-2.04	-1.18	-0.51	0.77	0.48	-1.43	-0.58	0.10	1.37	Large ab DIF	0.48	-1.43	-0.58	0.10	1.37	Large ab DIF
4	1.01	-1.80	-0.65	-0.20	0.86	1.01	-1.80	-0.65	-0.20	0.86		0.33	-1.80	-0.65	-0.20	0.86	Large a DIF
5	1.21	-2.03	-1.06	-0.48	0.70	1.21	-2.03	-1.06	-0.48	0.70		1.21	-1.38	-0.42	0.17	1.35	Large b DIF
6	0.93	-2.53	-1.24	-0.50	1.03	0.93	-2.53	-1.24	-0.50	1.03		0.93	-2.53	-1.24	-0.50	1.03	
7	0.71	-2.98	-1.62	-0.83	0.62	0.18	-2.12	-0.76	0.03	1.48	Large ab DIF	0.18	-2.12	-0.76	0.03	1.48	
8	1.33	-1.48	-0.53	-0.09	0.96	1.33	-1.48	-0.53	-0.09	0.96		1.33	-1.48	-0.53	-0.09	0.96	
9	1.61	-1.85	-0.92	-0.35	0.84	1.61	-1.85	-0.92	-0.35	0.84		1.61	-1.85	-0.92	-0.35	0.84	
10	1.61	-1.81	-1.07	-0.56	0.39	1.61	-1.81	-1.07	-0.56	0.39		1.61	-1.81	-1.07	-0.56	0.39	
11	1.43	-1.85	-0.89	-0.33	0.88	0.90	-1.55	-0.59	-0.02	1.18	Small ab DIF	0.90	-1.55	-0.59	-0.02	1.18	Small ab DIF
12	1.33	-1.89	-0.79	-0.33	0.70	0.66	-1.89	-0.79	-0.33	0.70	Small a DIF	0.66	-1.89	-0.79	-0.33	0.70	Small a DIF
13	1.61	-1.40	-0.45	-0.04	1.01	1.61	-1.11	-0.15	0.26	1.31	Small b DIF	1.61	-1.11	-0.15	0.26	1.31	Small b DIF
14	1.55	-1.61	-0.64	-0.17	0.95	1.55	-1.61	-0.64	-0.17	0.95		1.55	-1.61	-0.64	-0.17	0.95	
15	1.69	-1.53	-0.72	-0.31	0.56	0.90	-1.24	-0.43	-0.02	0.85	Small ab DIF	0.90	-1.24	-0.43	-0.02	0.85	Small ab DIF

Second, in each of the groups, the item fit statistics were excellent. Based on the $S\text{-}\chi^2$ statistic, there were only two items (Item 10 for Group 1; Item 9 for Group 2) that showed significant difference between the observed and modeled response frequencies ($p < .05$). This strongly suggests that the GRM fit the large majority of the items well. The two items in which DIF were found were not consistent across groups and may also be a result of the simulated DIF.

Third, we found little evidence for multidimensionality given small standardized local dependence $\chi^2(\text{LD-}\chi^2)$ in all three groups. There were two instances where the item pairs had values of 3.2 and 3.4, but the majority of the values were low (≤ 3.0). Based on these statistics, we conclude that a unidimensional GRM is appropriate.

DIF Analysis. We follow the steps outlined in Table 7 to examine DIF. As shown in Table 11, we initially fit Model 1 where item parameters are fully invariant across all three groups. It is noted that while the M_2 statistic revealed that the overall model was significantly different from the observed data, the RMSEA value was still low. Again, this suggests that the M_2 statistic may have limited usefulness for DIF analysis, and it needs to be complemented with other model-fit statistics.

At the next step, we sought to identify anchor items with Model 2 by using the two-step estimation procedure. The two-step procedure allows for all items to be tested for DIF. The identification of DIF was based on the use of two contrasts: comparing item parameters of the reference group to both the focal groups (Contrast 1) and comparing item parameters between the two focal groups (Contrast 2). We found that there were eight items that exhibited DIF between the reference group and the two focal groups (Column of Contrast 1). In addition, there were four items that exhibited DIF between the two focal groups: Group 2 and Group 3 (Column of Contrast 2). We note that it is possible for an overlap so that an item can exhibit DIF across both contrasts indicating that item parameters vary between all three groups. In this case, Items 4 and 12 exhibited DIF across all three groups.

Because the two-step procedure has high power but high Type I error rates, we still needed to test for DIF in the DIF items found in Model 2. At the next step (Model 3), we used the non-DIF items (items that did not show DIF on both DIF contrasts) in Model 2 as anchor items; namely, Items 1, 2, 8, 9, and 10. We continued to find DIF in all the items with the exception of Item 14. As such, in Model 4 we included Item 14 as an anchor. The same set of DIF items was found, and this confirmed our set of anchor items.

The final model, Model 5, was run with different contrasts to determine the presence of DIF between each focal groups and the reference group. Specifically, Contrast 3 tests whether there is DIF between the reference group and the first focal group (Groups 1 and 2); Contrast 4 tests whether there is DIF between the reference group and the second focal group (Groups 1 and 3). This is necessary because Contrast 1, which tests whether there is a difference between Group 1 and Groups 2 and 3 combined, while useful for assessing the presence of DIF, is not sensitive to whether DIF is attributable to a specific group. Further, Contrast 2, although helpful in determining DIF between Groups 2 and 3, does not address the commonly asked substantive question of whether items have DIF between the reference group and each focal group. Moreover, the procedure implemented for testing DIF between focal groups is usually less accurate as illustrated, where two items were correctly identified as having DIF between Groups 2 and 3 but two items are incorrectly identified as having DIF between Groups 2 and 3. Therefore, Contrasts 3 and 4 are important to specify. Based on these new contrasts, we found that DIF was accurately identified in all cases between the reference group and each focal group. We found that the theta mean levels estimated were also accurate; there was no difference between Groups 1 and 2 but Group 3 had a significantly lower level.

Effect Size Estimation. Using the latent trait estimation approach, we compare Model TB and Model T, which corresponds to Model 1 and Model 5 in Table 11, respectively. In Model TB, the focal group means for Groups 2 and 3 are estimated at $-.13$ and $-.49$, respectively. In Model T, the focal group

Table 11. Illustration 2: three group differential item functioning (DIF) analysis for GRM (graded response model).

Model	M2 prob	M2 Root Mean Square Error of Approximation (RMSEA)	Akaike Information Criterion (AIC)	Bayesian Information Criterion (BIC)	Items with significant DIF		Anchor Items	Theta Mean (SE)	Comment
					Contrast	Items with significant DIF			
1	0.0001	0.01	5,7672.99	5,8092.74	—	—	—	G1: .00 (—) G2: -.13 (.06) G3: -.49 (.06)	Full measurement equivalence across items
2	0.0001	0.00	5,6397.40	5,7592.88	3, 4, 7, 11, 12, 13, 14, 15	4*, 5*, 6, 12	—	G1: .00 (—) G2: -.13 (—) G3: -.49 (—)	Two-step procedure used to test for DIF. Step 1: Model 1 is estimated to obtain Groups 2 and 3 theta means and SDs; Step 2: Mean and SD values of theta values for Groups 2 and 3 are used—all items are freely estimated across groups and tested for DIF
3	0.0001	0.00	5,6338.04	5,7289.11	3, 4, 5, 7, 11, 12, 13, 15	4*, 5*, 6, 12,	1†, 2†, 8†, 9†, 10†	G1: .00 (—) G2: -.01 (.07) G3: -.35 (.06)	DIF analysis conducted by using all non-DIF items in Model 2 as anchor items
4	0.0001	0.00	5,6330.15	5,7228.08	3, 4, 5, 7, 11, 12, 13, 15	4*, 5*, 6, 12	1†, 2†, 8†, 9†, 10†, 14†	G1: .00 (—) G2: -.00 (.07) G3: -.35 (.06)	DIF analysis conducted by using all non-DIF items in Model 3 as anchor items
Model	M2 prob	M2 RMSEA	AIC	BIC	Contrast 3: 1, -1, 0	Contrast 4: 1, 0, -1	Anchor Items	Theta Mean (SE)	Comment
5	0.0001	0.00	5,6330.15	5,7228.08	3*, 7*, 11*, 12*, 13*, 15*	3*, 4*, 5*, 7*, 11*, 12*, 13*, 15*	1†, 2†, 8†, 9†, 10†, 14†	G1: .00 (—) G2: -.00 (.07) G3: -.35 (.06)	DIF analysis conducted by using all non-DIF items in Model 3 as anchor items and using different contrasts. Item 6 was not found to have DIF for Contrasts 3 and 4.

Note. * represents items that are correctly identified as having DIF; † represents items that are correctly identified as having no DIF (anchor items)

Table 12. Illustration 3: simulated item and theta parameters.

	Reference (Intercept = 0)			X1 (beta = −.34)				X2 (beta = .58)		
	γ	$\underline{a}$	$\underline{b}$	$\underline{a}$	$\underline{b}$	Type of DIF		$\underline{a}$	$\underline{b}$	Type of DIF
1	0.9	−0.26	2.06	−0.29	2.06	−0.29		2.06	−0.29	
2	0.66	−0.06	0.88	−0.09	0.88	−0.09		0.88	−0.09	
3	0.83	−0.42	1.49	−0.51	1.49	−1.01	Large b DIF	1.49	−0.31	Small b DIF
4	0.71	−0.14	1.01	−0.20	1.01	−0.20		1.01	−0.20	
5	0.77	−0.37	1.21	−0.48	1.21	−0.48		1.21	−0.48	
6	0.68	−0.34	0.93	−0.50	0.93	−0.50		0.93	−0.50	
7	0.58	−0.48	0.71	−0.83	0.71	−1.33	Large b DIF	0.71	−0.63	Small b DIF
8	0.8	−0.07	1.33	−0.09	1.33	−0.09		1.33	0.21	Small b DIF
9	0.85	−0.30	1.61	−0.35	1.61	−0.35		1.61	−0.35	
10	0.85	−0.48	1.61	−0.56	1.61	−0.56		1.61	−0.56	
11	0.82	−0.27	1.43	−0.33	1.43	−0.58	Small b DIF	1.43	−0.33	
12	0.8	−0.26	1.33	−0.33	1.33	−0.33		1.33	0.27	Large b DIF
13	0.85	−0.03	1.61	−0.04	1.61	−0.04		1.61	0.46	Large b DIF
14	0.84	−0.14	1.55	−0.17	1.55	−0.17		1.55	−0.17	
15	0.86	−0.27	1.69	−0.31	1.69	−0.56	Small b DIF	1.69	−0.31	

means for groups 2 and 3 are estimated at .00 and −.35, respectively. Therefore, the magnitude of the total bias, based on the difference between Models TB and T, for the two focal groups are .13 and .14, respectively. This represents a level of bias that is quite small. Nevertheless, for the first focal group (Group 2), not accounting for the bias can lead to an incorrect statistical inference that there is a group difference between this focal group and the reference group when there is none. This implies that not accounting for bias can lead to an incorrect substantive inference between the reference group and the first focal group when conducting a statistical test without accounting for DIF.

Illustration 3: Covariate Approach to 3PLM IRT DIF Analysis

In this final illustration, we show how to implement the IRT-C procedure in Latent GOLD. This procedure allows for a general way to assess DIF on multiple group characteristics. Because the software Latent GOLD does not produce multiple IRT fit statistics, we use IRTPRO to check for unidimensionality and model-data fit before testing for DIF.

Simulating Data. We simulated the association between theta and covariates with the following equation, where $\theta = 0.00 - .20(X1) + .30(X2) + e$, X1 is a dichotomous variable (0 or 1) representing group differences (e.g., male vs. female; Black vs. White); X2 is a standardized continuous variable (e.g., age, income); and the R square value simulated was .30. The expected value of the coefficients of X1 and X2 is −.34 and .58, respectively, when the theta value is standardized (this is typically done in IRT estimation—and in Latent GOLD—to resolve scale indeterminacy).

We adapted item parameters from Illustration 1 where we added pseudo-guessing *c* parameters such that the first set of five items has *c* parameters of .15, the second set of five items has *c* parameters of .20, and the last set of five items has *c* parameters of .25.

We simulated DIF such that being in the focal group X1 leads to greater easiness (with different magnitudes) in responding to certain items as shown in Table 12. In addition, being one standard deviation higher on X2 leads to greater difficulty in responding to specific items. As such, the DIF magnitude of .20 reflects a DIF value for X2 values one standard deviation above the mean; by extension, the DIF magnitude would be .40 for X2 values two standard deviations higher. A total of 5,000 simulees were generated.

Fitting a Unidimensional 3PLM. Initially, we used IRTPRO to fit a 3PLM to all the data to assess the overall model-data fit. In other words, we did not create separate groups for the covariates. We note that for 3PLM estimation, it is helpful to include a prior beta distribution in order to provide accurate estimates of the c -parameters.¹ We found that the model had reasonably low RMSEA values of .01 but it was significant ($p < .05$). All the item-level $S-\chi^2$ statistics were not significant, indicating that the modeled frequencies were similar to the observed frequencies. A quick comparison of overall fit for the 3PLM (RMSEA = .01; AIC = 86,218.74; BIC = 86,512.01) was done with a 2PLM (RMSEA = .02; AIC = 86,248.91; BIC = 86,444.43) and 1PLM (RMSEA = .03; AIC = 86,685.27; BIC = 86,789.55). We found that the 3PLM had the lowest absolute fit and relative fit using the AIC; however, the relative fit using BIC showed that the 2PLM fit better. A comparison of log-likelihoods showed in a significant difference between the 2PL and the 3PL models ($p < .05$). As such, we chose to go with the 3PLM because of the better absolute fit, lower AIC, and significantly better fit based on the log-likelihood statistic. When we examined the standardized LD- χ^2 statistic, we found that although there were 3 pairs of items that had large standardized LD- χ^2 of magnitudes 4.2, 5.8, and 33.8, all other item pairs had low standardized LD- χ^2 values demonstrating general unidimensionality.

DIF Analysis. In Table 13, we show the iterative models used to ascertain DIF on the items for a specific group characteristic (i.e., covariate X1 or X2). The steps for this procedure are outlined in Table 7. In the initial Model 1 we fit a fully constrained baseline model and used the item-covariate BVR to flag the most likely item that has DIF on a group characteristic. Item 13 on group characteristic X2 had the largest BVR value of 201.13. In the subsequent Model 2, we found that this DIF was significant. In addition, we found that item 12 most likely had DIF on group characteristic X2, with the largest BVR value of 154.96. Similar steps were repeated until the flagged item did not display significant DIF. As seen in Model 11, Item 5 on group characteristic X2 did not show significant DIF. As such, the final model chosen is the prior Model 10, where DIF of Item 5 on group characteristic X2 was not modeled. Coincidentally, we find that Model 10 also had the lowest information criteria as compared to the other fitted models, strengthening the argument that this is the best model. Importantly, all the DIF that was simulated was correctly identified through this procedure.

Effect Size. In the initial model (Model 1), where DIF was not accounted for, the latent trait estimates reflects true differences and bias introduced by DIF (Model TB). For Model TB, the coefficients associated with X1 and X2 were $-.26$ and $.49$, respectively. In the final model (Model 11) where DIF was accounted for, we expect that only true differences are estimated (Model T). For Model T, the coefficients for X1 and X2 were $-.43$ and $.62$, respectively. As such the magnitude of bias for X1 was $|-.26 - (-.43)| = .17$ and for X2 was $|.49 - .62| = .13$. The magnitudes of bias were therefore small. With regard to effects on statistical testing, we found that significance testing of the group characteristics (from the reference group) yielded the same results. In line with the simulated parameters, individuals belonging to group X1 have significantly lower trait values and individuals with higher levels of X2 have significantly higher trait values.

Summary of Recommendations and Additional Issues

We present a summary of our key recommendations and also note additional issues that would be helpful for researchers seeking to use IRT DIF analysis to be mindful of.

Table 13. Illustration 3: covariate approach differential item functioning (DIF) analysis for 3 parameter logistic model (PLM).

Model	Akaike Information Criterion (AIC)	Bayesian Information Criterion (BIC)	Akaike's Information Criterion (CAIC)	Freely Estimated Items (identified DIF items)	p Value of Previous Bivariate Residual (BVR) Item	Highest BVR Item-Covariate Pair	B Parameters (SE)	Comment
1	85,518.51	85,824.82	85,871.82	—	—	13-X2 (201.13)	X1: -.26 (.03) X2: .49 (.02)	Full measurement equivalence across items
2	85,135.47	85,448.29	85,496.29	13-X2*	13-X2 ($p < .001$)	12-X2 (154.96)	X1: -.27 (.04) X2: .53 (.02)	
3	84,802.22	85,121.56	85,170.56	13-X2*, 12-X2*	12-X2 ($p < .001$)	3-X1 (92.27)	X1: -.28 (.04) X2: .56 (.02)	
4	84,660.15	84,986.01	85,036.01	13-X2*, 12-X2*, 3-X1*	3-X1 ($p < .001$)	7-X1 (35.08)	X1: -.33 (.03) X2: .56 (.02)	
5	84,623.12	84,955.49	85,006.49	13-X2*, 12-X2*, 3-X1*, 7-X1*	7-X1 ($p < .001$)	8-X2 (28.07)	X1: -.34 (.03) X2: .56 (.02)	
6	84,514.89	84,853.79	84,905.79	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*	8-X2 ($p < .001$)	3-X2 (28.03)	X1: -.34 (.04) X2: .59 (.02)	
7	84,450.22	84,795.63	84,848.63	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*, 3-X2*	3-X2 ($p < .001$)	15-X1 (27.40)	X1: -.34 (.03) X2: .61 (.02)	
8	84,397.73	84,749.65	84,803.65	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*, 3-X2*, 15-X1*	15-X1 ($p < .001$)	11-X1 (28.58)	X1: -.38 (.04) X2: .61 (.02)	
9	84,342.18	84,700.63	84,755.63	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*, 3-X2*, 15-X1*, 11-X1*	11-X1 ($p < .001$)	7-X2 (8.37)	X1: -.43 (.04) X2: .61 (.02)	
10	84,331.25	84,696.22	84,752.22	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*, 3-X2*, 15-X1*, 11-X1*, 7-X2*	7-X2 ($p < .01$)	5-X2 (1.14)	X1: -.43 (.04) X2: .62 (.02)	
11	84,331.26	84,702.74	84,759.74	13-X2*, 12-X2*, 3-X1*, 7-X1*, 8-X2*, 3-X2*, 15-X1*, 11-X1*, 7-X2*, 5-X2	5-X2 (ns)	9-X2 (1.02)	X1: -.43 (.03) X2: .62 (.02)	Procedure ends because the flagged item does not have DIF. In addition, information criteria increased.

Note. * represents items that are correctly identified as having DIF.

Research Design

Sample Size. In general, larger sample sizes give rise to increased estimation accuracy. Past research with IRT has not focused on recommendations of sample sizes as IRT has historically been applied to educational tests where sample sizes are large. Recent research has examined the extent parameter-to-sample size ratios serve as a good rule of thumb for good item parameter recovery in the NRM (De Ayala & Sava-Bolesta, 1999; DeMars, 2003) but not for other IRT models typically used. We continue to recommend that a minimum sample size of 500 per group is sufficient for IRT DIF analysis (Hulin et al., 1982). In general, the total sample size that is dependent on the groups may be more important. For example, a total sample size of 1,000 for a two-group IRT DIF analysis can yield sufficient power ($\sim .80$) (e.g., Woods et al., 2013). However, sample sizes need not be equal as it has been shown that a sample size of 750 in the reference group and 250 in the focal group(s) has also been found to have acceptable power ($\sim .80$) (Woods et al., 2013). For the IRT-C procedure, where multiple covariates are used, we also recommend the use of a minimum of 500 participants per covariate level. If, for instance, two covariates X1 (two levels: male vs. female) and X2 (three levels: White, Black, Asian) are used, it would require a minimum of $2 \times 3 \times 500 = 3,000$ participants. A continuous covariate would be equivalent to five levels. Therefore if two covariates X1 (two levels: male vs. female) and X2 (continuous: age) are used, it would require a minimum of $2 \times 5 \times 500 = 5,000$ participants.

Numbers of Items. There are usually no fixed rules of thumb on the minimum numbers of items needed to apply IRT, although similar to CFA, at least 3 items are needed for a just-identified model. As found in our review, studies have applied IRT to as few as 4 items. We propose that a minimum of 4 or more items can be used, although it is usually preferable to have 10 items or more (as larger values are more commonly simulated in testing IRT DIF statistics/procedures and their properties are well known) (e.g. Meade, Lautenschlager, & Johnson, 2007; Stark et al., 2004; Woods et al., 2012). In addition, when assessing IRT DIF without preknowledge of anchor items, having more items provides a greater chance that more items are equivalent.

Type of IRT Model. We refer individuals to Table 4 for the different types of IRT models one can specify and how they are generally applied. As mentioned earlier in the article, the choice of the IRT model often depends on (a) the number of item parameters modeled, (b) how the categorical responses are treated (nominal or ordered), and (c) the underlying response process (monotonic or ideal point models).

We propose that when there is contention on which IRT model is appropriate, it is helpful to compare model-data fit between competing models. For example, one can test whether a 1PLM that assumes that all item discriminations are all equal is preferable to a 2PLM, which assumes that item discriminations may differ. Further, we encourage researchers to report the formula and parameterization of IRT model used so that the reported item parameter values can be clearly understood.

Critical Issues to Check and Report. Because the estimation accuracy is dependent on the combination of sample size, numbers of items and the type of IRT model, it is imperative that researchers check for how well the final IRT model fits. This includes checking the standard errors (*SEs*) and the magnitudes of the item parameters.

Researchers should check the *SEs* of item parameters to ensure that the IRT model is well estimated. Lower values of *SEs* for item parameters (item discrimination, location/thresholds, guessing) indicate better fit. From experience, for the logistic parameterization (1PLM, 2PLM, 3PLM, and GRM), the average standard error values for item parameters indicating poor to excellent fit are:

- $>.50$ indicate poor estimates,
- $.35$ to $.50$ indicate moderate estimates,

- .20 to .35 indicate good estimates,
- <.20 indicate excellent estimates.

Where the *SEs* are large, it suggests that a larger sample size is needed. We note that the *SEs* can be used in the review process to demonstrate that an IRT model is well estimated. For instance, despite a small sample size (e.g., 300 participants), the average *SEs* when shown to be in the range of .20 to .30 for the item parameters would demonstrate that the IRT model is well estimated.

Another critical issue to ensure is that the item parameters are reasonable in size. Specifically, for the logistic parameterizations (1PLM, 2PLM, 3PLM, and GRM), multiple *b*-parameters that have exceedingly large magnitudes (e.g., >3) would be problematic. This is because the latent trait is scaled as $N(0,1)$, and the item locations are scaled relative to this latent trait distribution. Large magnitudes suggest that the sample used may be inappropriate and this is usually associated with large *SEs*. In such instances, we recommend that researchers use a different sample for item estimation.

DIF Analysis

In general, we recommend that two or more IRT fit statistics be used and reported when testing unidimensionality and model-data fit (see Table 5). This is because some of the developed fit-statistics have not been well validated (e.g., M2 statistic) and the use of multiple fit statistics can augment potential weaknesses. For example, in IRTPRO, one can report the overall fit using the M2 statistic, item-level fit using the $S\text{-}\chi^2$, and unidimensionality using the standardized LD χ^2 . There are also other combinations depending on what a researcher uses for analysis. For instance, one can report the results of the PCA using statistical software such as SPSS and also report the average BVRs in Latent GOLD.

The general procedures for testing DIF are shown in Tables 6 and 7. It is similarly helpful to rely on two indicators when testing for DIF. For example, one can examine and report the information criteria of competing models and also the significance from the Wald χ^2 statistic for the tested items. One important issue is that in testing for DIF when anchor items are not known a priori, it is important that the proportion of DIF detected is not more than 50%. In these instances, DIF items may mask as non-DIF items in the analysis; conversely, non-DIF items may appear as having DIF. This is because true DIF items could be inadvertently used to link the metrics between groups and it can lead to detecting true non-DIF items as having DIF (because the wrong linking is used). A possible cause of this is that the scale is too multidimensional and subdimensions should be analyzed separately for DIF. Another possible reason for this is that the wrong type of IRT model has been applied to the data.

With regard to DIF effect size estimation procedures, we propose that the heuristic approach of estimating DIF effect size is usually sufficient for most purposes. This is because researchers usually seek to determine the DIF effect size between the groups across the latent trait distribution as the goal is to determine overall mean differences between groups. When one seeks to assess the DIF effect size on a specific latent trait value or interval—often for the purposes of demonstrating that a cutoff value does not have appreciable levels of DIF—then the expected score effect size approach is preferable.

Finally, although for convenience we have focused on the use of IRTPRO and Latent GOLD, we note that there are additional software that can be used for IRT estimation and IRT DIF analysis (see Tables 5 and 6). There are also a variety of free packages for users of R (<http://cran.r-project.org/web/views/Psychometrics.html>). A caveat is that many of these software and R packages do not provide seamless reporting of the necessary model-data fit, DIF indices, and effect size estimates.

ME/DIF and Substantive Interpretations

One common substantive interpretation of finding CFA or IRT ME (or a lack of DIF) is that constructs are *interpreted* in the same manner across groups. We propose that this argument may be problematic

(see also Tay, Woo, & Vermunt, 2014). In making this argument, we are inferring from between-individual scores to within-individual cognitions. With ME analysis, we use between-individual scores to assess whether construct indicators are related to the latent trait in the same manner across groups. It is difficult to infer whether any single individual in a given group thinks about the construct in the same way. However, we can infer that observed scores are comparable at the group level if there is no ME. This is usually the object of ME analysis—to show that observed scores are comparable so that substantive interpretations can be made on whether groups are equal or different on a construct of interest.

When there is a lack of ME or when DIF is detected, researchers need to be careful when inferring the cause of DIF. This is because there may be other confounds not accounted for. For example, DIF between two nations may not be due to culture. It could be attributable to differences in samples (e.g., university students vs. workers) or language (e.g., Mandarin Chinese vs. English). In order to infer the causes of DIF, we recommend that more rigorous designs be used. For example, one can examine DIF among three groups of university students using (a) an English version administered in the United States, (b) an English version administered in China, and (c) a Chinese version administered in China to determine whether DIF is primarily attributable to culture, language, or both. See Robert, Lee, and Chan (2006) for an illustration.

Future Research and Conclusion

The focus on how to practically assess IRT DIF has limited our consideration of several developing areas of DIF research. We note that the procedures and statistics for DIF analysis is an ongoing area of research and, in our opinion, more active than CFA ME research. As such, there are other DIF indices/procedures that are not included here. Researchers continue to propose variations of indices for assessing DIF along with new procedures. For instance, different variations of MH statistics continue to be developed and implemented (Gonzalez, Padilla, Hidalgo, Gomez-Benito, & Benitez, 2011). Different forms of BVRs have been tested to determine which more effectively detects potential DIF (Tay et al., 2013). New procedures are being developed to identify anchor items associated with the IRT likelihood ratio test (Meade & Wright, 2012) and the MH procedure (Gonzalez et al., 2011). While we have not focused on all the latest variations of IRT ME/DIF analysis, this review suggests that an area of research that appears to be particularly needed is establishing the best cutoff criteria for model-data fit.

While research suggests that the M_2 -statistic is statistically optimal compared to other model-data fit indices such as the adjusted χ^2 -statistic (Drasgow et al., 1995), less is known about the practical cutoff values that indicate good fit. This is unlike CFA ME analysis that relies on traditional SEM indicators of model-data fit, and more research has been conducted on determining which indices are useful for detecting nonequivalence (e.g., Cheung & Rensvold, 2002). To overcome the problem of using rules of thumb that may not generalize to all scenarios (Nye & Drasgow, 2010), one possibility for assessing overall model-data fit is to use a parametric bootstrap analysis and examine the associated p values (Vermunt & Magidson, 2005). For current practical purposes, however, the recommendations of using multiple criteria to test unidimensionality and model-data fit should suffice (see Table 5).

In conclusion, our guide to IRT ME/DIF analysis seeks to integrate key DIF indices/procedures and provides best practices in an accessible manner to organizational researchers and more generally, to other fields such as psychology and education that require testing for IRT DIF. We believe that in this article, we have outlined two convenient yet effective ways of assessing IRT DIF. We hope that researchers can use this framework for conducting IRT ME/DIF analysis. Further, we expect that this article can serve to stimulate the application of IRT ME/DIF analysis to organizational science and further basic research in IRT ME/DIF analysis.

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Note

1. $m = 20$; $p = .20$; $\text{Alpha} = mp + 1 = 5$; $\text{Beta} = m(1 - p) + 1 = 17$.

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